MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

RECEIVER, LINE, DIGITAL, KEY CODE BDM

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic line receiver module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.446 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be BDM.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/196-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

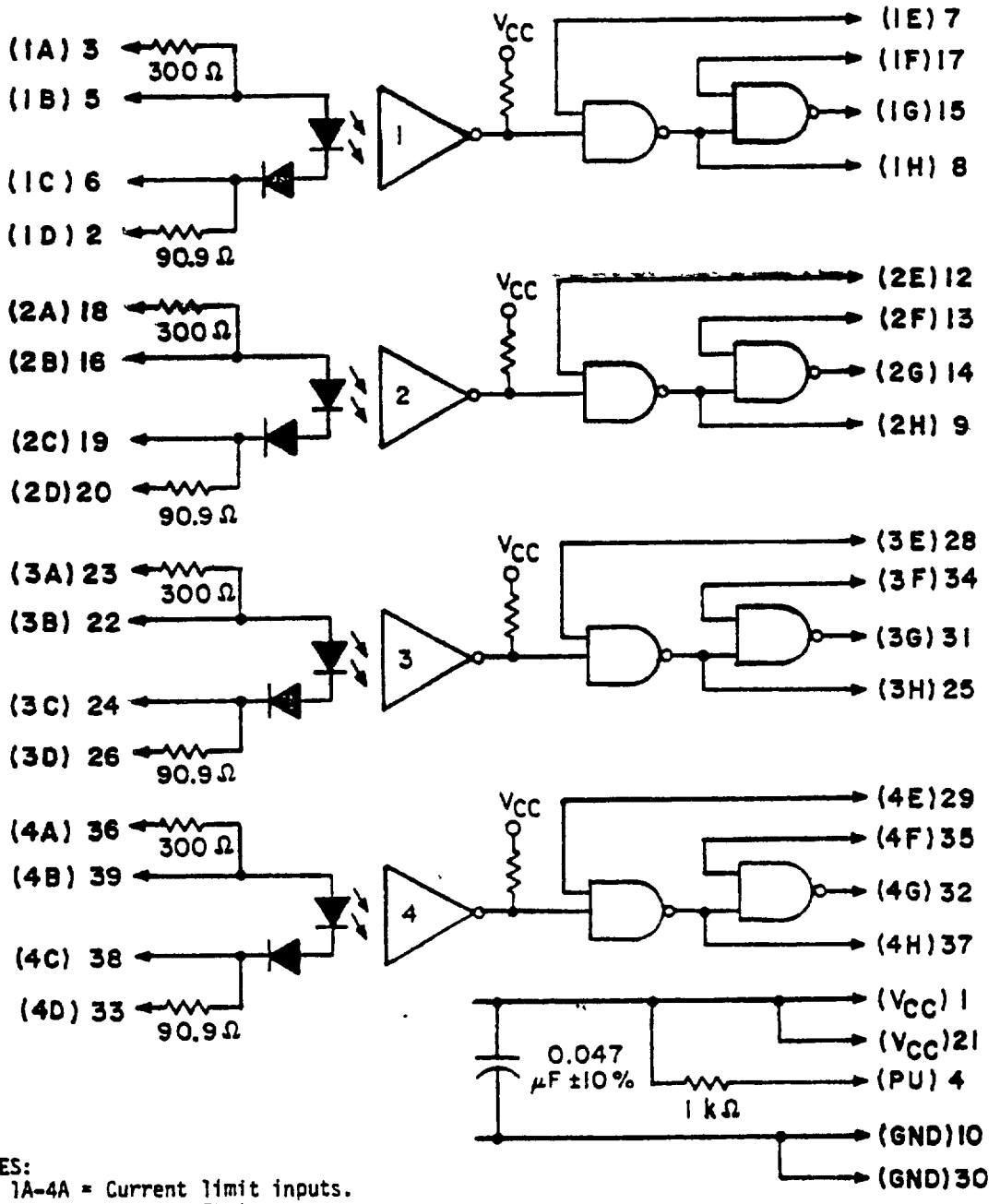
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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NOTES:
1. 1A-4A = Current limit inputs.
2. 1B-4B = Direct LED inputs.
3. 1C-4C = A and B signal return.
4. 1D-4D = Programmable terminator input.
5. 1E-4E = Inhibit inputs.
6. 1F-4F = Inhibit inputs.
7. 1G-4G = Outputs
8. 1H-4H = Complementary outputs.

* FIGURE 1. BDM module function diagram, line receiver.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A6	Isolation	III	X	10	∞	1	∞	10	∞	1	∞	MΩ
B1-B4	Termination resistor voltage	IV	Q	9.31	9.60	9.22	9.70	9.21	9.71	9.20	9.72	V
C1-C4	LED forward voltage, high level	IV	Q	2.20	2.45	2.20	2.45	2.00	2.55	2.00	2.55	V
C5-C8	LED forward voltage, low level	IV	Q	1.3	2.3	1.3	2.3	1.2	2.4	1.2	2.4	V
D1-D4	LED reverse breakdown voltage (measuring leakage current)	V	Q	0.0	0.5	0	5	0	1	0	8	µA
E1-E8	Common mode voltage, low level	VI	S	20	400	20	400	20	400	20	400	mV
E9-E16	Common mode voltage, high level	VI	S	2.4	5.0	2.4	5.0	2.4	5.0	2.4	5.0	V
F1-F8	V_{OUT} (low loaded)	VII	S	50	500	50	500	50	500	50	500	mV
G1-G8	Input breakdown current	VIII	S	0	1	0	1	0	1	0	1	mA
H1-H4	Output short circuit current	IX	X	-20	-55	-20	-55	-20	-55	-20	-55	mA
H5-H8	Output short circuit current	IX	X	-20.0	-56.7	-20.0	-56.7	-20.0	-56.7	-20.0	-56.7	mA
J1-J8	Input current, high level	X	X	0	40	0	40	0	40	0	40	µA
K1-K8	Input current, low level	XI	X	-0.70	-1.73	-0.70	-1.73	-0.70	-1.73	-0.70	-1.73	mA
L1-L8	Functional test	XII	X	←---Test vectors must produce output vectors as specified in table XII.---→								
M1-M4	Input threshold voltage (high to low)	XIII	X	0.8	2.0	0.8	2.0	0.8	2.0	0.8	2.0	V
N1-N4	Input threshold voltage (low to high)	XIV	X	0.6	2.0	0.6	2.0	0.6	2.0	0.6	2.0	V
P1-P8	Output voltage, low level	XV	X	20	400	20	400	20	400	20	400	mV
Q1-Q8	Output voltage, high level	XVI	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
R1,R2	Continuity	XVII	S	0	1	0	1	0	1	0	1	Ω
R3-R6	Resistance	XVII	X	293.7	306.3	281.7	318.3	291.7	308.1	279.9	320.1	Ω
R7-R10	Resistance	XVII	X	90.0	91.8	88.0	93.8	89.7	92.1	87.9	93.9	Ω

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
R11	Resistance	XVII	X	900	1,100	890	1,110	890	1,110	880	1,120	Ω
S1	Power supply current, G and H outputs high	XVIII	X	20.0	78.8	20.0	78.8	20.0	78.8	20.0	78.8	mA
S2	Power supply current, G outputs low, H outputs high	XVIII	X	55	125	55	125	55	125	55	125	mA
T1-T8	Input clamp diode voltage	XIX	S	-0.4	-1.5	-0.4	-1.5	-0.3	-1.5	-0.3	-1.5	V
U1-U8	Transition time (t_{TLH}) C_L = 50 pF	XX	S	1	13	1	13	1	13	1	13	ns
U9-U16	Transition time (t_{TLH}) C_L = 150 pF	XX	Q	3	21	3	21	3	21	3	21	ns
V1-V8	Transition time (t_{THL}) C_L = 50 pF	XX	S	1	10	1	10	1	10	1	10	ns
V9-V16	Transition time (t_{THL}) C_L = 150 pF	XX	Q	1	17	1	17	1	17	1	17	ns
W1-W4	Propagation delay time (t_{PLH}), C_L = 50 pF	XX	S	10	125	10	125	10	125	10	125	ns
W5-W8	Propagation delay time (t_{PLH}), C_L = 50 pF	XX	S	13	145	13	145	13	145	13	145	ns
W9-W12	Propagation delay time (t_{PLH}), C_L = 150 pF	XX	Q	15	145	15	145	15	145	15	145	ns
W13-W16	Propagation delay time (t_{PLH}), C_L = 150 pF	XX	Q	20	175	20	175	20	175	20	175	ns
X1-X4	Propagation delay time (t_{PHL}), C_L = 50 pF	XX	S	10	120	10	120	10	120	10	120	ns

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
X5-X8	Propagation delay time (t_{PHL}), C_L = 50 pF	XX	S	13	145	13	145	13	145	13	145	ns
X9-X12	Propagation delay time (t_{PHL}), C_L = 150 pF	XX	Q	15	135	15	135	15	135	15	135	ns
X13-X16	Propagation delay time (t_{PHL}), C_L = 150 pF	XX	Q	20	177	20	177	20	177	20	177	ns
Y1-Y4	Propagation delay time (t_{PLH}), C_L = 50 pF	XXI	S	3	25	3	25	3	25	3	25	ns
Y5-Y8	Propagation delay time (t_{PLH}), C_L = 50 pF	XXI	S	6	45	6	45	6	45	6	45	ns
Y9-Y12	Propagation delay time (t_{PLH}), C_L = 50 pF	XXI	S	3	25	3	25	3	25	3	25	ns
Y13-Y16	Propagation delay time (t_{PLH}), C_L = 150 pF	XXI	Q	5	42	5	42	5	42	5	42	ns
Y17-Y20	Propagation delay time (t_{PLH}), C_L = 150 pF	XXI	Q	10	75	10	75	10	75	10	75	ns
Y21-Y24	Propagation delay time (t_{PLH}), C_L = 150 pF	XXI	Q	5	42	5	42	5	42	5	42	ns
Z1-Z4	Propagation delay time (t_{PHL}), C_L = 50 pF	XXI	S	3	27	3	27	3	27	3	27	ns
Z5-Z8	Propagation delay time (t_{PHL}), C_L = 50 pF	XXI	S	6	52	6	52	6	52	6	52	ns
Z9-Z12	Propagation delay time (t_{PHL}), C_L = 50 pF	XXI	S	3	20	3	20	3	20	3	20	ns
Z13-Z16	Propagation delay time (t_{PHL}), C_L = 150 pF	XXI	Q	5	40	5	40	5	40	5	40	ns

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See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
Z17-Z20	Propagation delay time (t_{PHL}), C_L = 150 pF	XXI	Q	10	82	10	82	10	82	10	82	ns
Z21-Z24	Propagation delay time (t_{PHL}), C_L = 150 pF	XXI	Q	5	33	5	33	5	33	5	33	ns
AA1-AA4	Common mode transient immunity, high level	XXII	Q	0	400	0	400	0	400	0	400	mV
BB1-BB4	Common mode transient immunity, low level	XXIII	Q	0	400	0	400	0	400	0	400	mV
CC1-CC16	Output crosstalk, high level	XXIV	Q	0	400	0	400	0	400	0	400	mV
DD1-DD16	Output crosstalk, low level	XXV	Q	0	400	0	400	0	400	0	400	mV
EE	Power supply transient amplitude	XXVI	Q	0	400	0	400	0	400	0	400	mV
FF	Power supply filter capacitance	XXVII	S	0.032	0.062	0.032	0.062	0.032	0.062	0.032	0.062	µF
GG1-GG8	Input capacitance	XXVIII	Q	2	15	2	15	2	15	2	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through GG shall be performed in any order with the following exceptions. If test conditions A, B, C, D, E, F, G, or H are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, E, F, G, H where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through GG in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions H through FF shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition FF, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests H through FF shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests H through FF shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Part number M28787/196-1 supersedes part number M28787/196. The two parts are interchangeable.

* 6.4 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life limits from table I.

Supply voltage	5.0 V dc ±10%
V_{CC} current (power dissipation)	V_{CC} = 5.5 V
G outputs at low level; H outputs at high level	125 mA maximum (688 mW)
G outputs at high level; H outputs at high level	78.8 mA maximum (433 mW)

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Input power dissipation (typical per circuit)	
A/C (V_{IH} = 6.5 V)	90 mW
D/C (V_{IH} = 6.5 V) pulsed at 50% dc	235 mW

When signals are applied to module inputs, power is dissipated in the resistor and diode networks at inputs A, B, C, and D as well as the output circuitry. Power ratings of the individual input components or total package power (2.5 watts at 60°C fin) must not be exceeded by any set of input conditions.

Differential low-level input voltage (V_{IL})	
B/C (direct LED input) for $I_{IL} \leq 0.25$ mA	1.2 V maximum
Differential low-level input current (I_{IL})	
A/C (current limit input)	0.25 mA maximum
B/C (direct LED input)	0.25 mA maximum
Differential high-level input voltage (V_{IH})	
B/C (direct LED input, I_{IH} = 20 mA)	2.55 V maximum
A/C (current limit input)	5 V minimum; 8 V maximum
A/C (current limit input) pulsed input ($t_{on} \leq 1$ ms at 50% duty cycle)	5 V minimum; 10.3 V maximum
Differential high-level input voltage (I_{IH})	
B/C (direct LED input)	8 mA minimum; 20 mA maximum
A/C (current limit input)	8 mA minimum; 20 mA maximum
B/C (direct LED input) pulsed input ($t_{on} \leq 1$ ms at 50% duty cycle)	40 mA maximum
A/C (current limit input) pulsed input $t_{on} \leq 1$ ms at 50% duty cycle)	29 mA maximum
Input voltage for inhibit inputs	
Low level (E input)	-0.5 V minimum; 0.6 V maximum
Low level (F input)	-0.5 V minimum; 0.8 maximum
High level (E and F inputs)	2 V minimum; V_{CC} maximum
Input current for inhibit inputs (E and F)	
Low level (V_{IL} = 0.0 V)	-1.73 mA maximum
(V_{IL} = 0.4 V)	-1.6 mA maximum
High level (V_{IH} = 2.4 V)	40 µA maximum
(V_{IH} = 5.5 V)	1 mA maximum
Output voltage	
G output, low level (I_{OL} = 16 mA)	0.4 V maximum
H output, low level (I_{OL} = 14.4 mA)	0.4 V maximum
G output, high level (I_{OH} = -400 µA)	2.4 V minimum
H output, high level (I_{OH} = -360 µA)	2.4 V minimum

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Output drive current, maximum capability	
G output, low level (V_{OL} = 0.4 V)	16 mA
H output, low level (V_{OL} = 0.4 V)	14.4 mA
G output, high level (V_{OH} = 2.4 V)	-400 µA
H output, high level (V_{OH} = 2.4 V)	-360 µA
V_{IR} (maximum reverse input voltage at I_{IR} = 8 µA)	50 V
Pin programmable input resistance networks (each circuit)	
A/B	300Ω ±2%, 1/8 W
C/D	90.9Ω ±1%, 1.0 W
Pullup resistance (between V_{CC} and module pin 4)	1 kΩ ±10%, 1/8 W
Terminated input impedance A/C (A input pin programmed to input D)	
V_{IH}	75Ω ±10%
V_{IL}	90.9Ω ±10%
Isolation resistance (each circuit input, A, B, C, D to any other point) (V_{IR} = 100 V)	1 MΩ minimum
Common mode voltage (each circuit input, A, B, C, D to any other point)	100 V
Short circuit output current	
G outputs	-20 mA minimum; -55 mA maximum
H outputs	-20 mA minimum; -56.7 mA maximum
Propagation delay time (t_{PLH})	
A/C to H	
(C_L = 50 pF)	10 ns minimum; 125 ns maximum
(C_L = 150 pF)	15 ns minimum; 145 ns maximum
A/C to G	
(C_L = 50 pF)	13 ns minimum; 145 ns maximum
(C_L = 150 pF)	20 ns minimum; 175 ns maximum
E to H	
(C_L = 50 pF)	3 ns minimum; 25 ns maximum
(C_L = 150 pF)	5 ns minimum; 42 ns maximum
E to G	
(C_L = 50 pF)	6 ns minimum; 45 ns maximum
(C_L = 150 pF)	10 ns minimum; 75 ns maximum
F to G	
(C_L = 50 pF)	3 ns minimum; 25 ns maximum
(C_L = 150 pF)	5 ns minimum; 42 ns maximum

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Propagation delay time (t_{PHL})	
A/C to H	
(C_L = 50 pF)	10 ns minimum; 120 ns maximum
(C_L = 150 pF)	15 ns minimum; 135 ns maximum
A/C to G	
(C_L = 50 pF)	13 ns minimum; 145 ns maximum
(C_L = 150 pF)	20 ns minimum; 177 ns maximum
E to H	
(C_L = 50 pF)	3 ns minimum; 27 ns maximum
(C_L = 150 pF)	5 ns minimum; 40 ns maximum
E to G	
(C_L = 50 pF)	6 ns minimum; 52 ns maximum
(C_L = 150 pF)	10 ns minimum; 82 ns maximum
F to G	
(C_L = 50 pF)	3 ns minimum; 20 ns maximum
(C_L = 150 pF)	5 ns minimum; 33 ns maximum
Common mode transient immunity (Transition time of a 10 V noise signal between inputs A, B, C, D, and module grounds, (10, 30) to cause output to exceed TTL logic thresholds, 0.8 V, 2.0 V)	
CMVH (Transition time of noise signal to change state when $I_{IH} \geq 8$ mA A/C)	100 ns minimum
CMVL (Transition time of noise signal to change state when I_{IL} = 0.0 mA A/C)	300 ns minimum

* 6.5 Module function. This module contains four line receiver circuits, each having a light emitting diode (LED) optically coupled to a photosensitive amplifier with a TTL compatible output. Each circuit has complementary logic outputs made available by driving 2-input NAND gates in cascade, the unused input to each NAND gate is brought out to module pins and can be used as inhibit inputs. A pullup resistor is also available at module pin 4. The module operates from a 5 V dc supply. The input network for each circuit is independent of all others and is pin programmable to provide:

a. Direct access to the LED for operation over a range of input voltages. (The current through the LED must be limited below its absolute maximum value to prevent its damage or degradation.)

b. An internal resistance to limit LED current and a termination resistance for 75-ohm line impedance. (When signals are applied to module inputs, power is dissipated in the resistor and diode networks of inputs A, B, C, D; power ratings of individual components or total package power (2.5 watts at 60°C fin) must not be exceeded by any set of input conditions.)

6.5.1 Input to output relationship. Each line receiver shall have the input to output relationship of table II for positive logic.

6.5.2 Parallel outputs. Paralleling of outputs is not permitted.

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6.5.3 Unused inhibit inputs. Inhibit inputs E and F shall be connected to high level when not in use. A pullup resistor is available at pin 4.

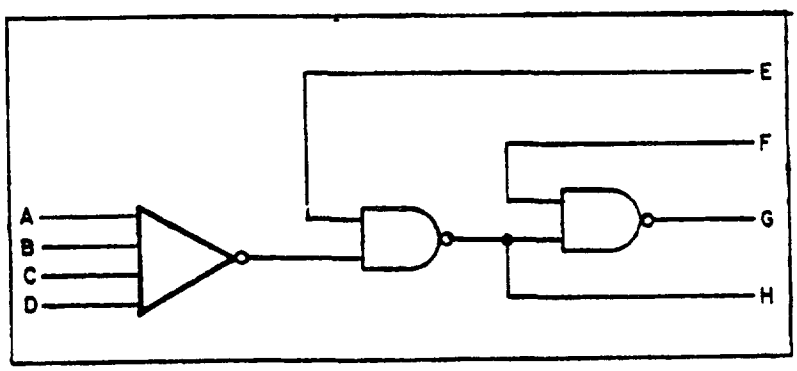
6.5.4 Input polarity. Observe polarity of input voltage. Positive input voltage at pins 5, 16, 22, 39, with respect to pins 6, 19, 24, and 38.

TABLE II. Input to output relationship. 1/ 2/

INPUTS						OUTPUTS	
A	B	C	D	E	F	G	H
	I_{IL}			H	H	H	L
	I_{IH}			H	H	L	H
	I_{IL}			L	H	L	H
	I_{IH}			L	H	L	H
	I_{IL}			H	L	H	L
	I_{IH}			H	L	H	H
	I_{IL}			L	L	H	H
	I_{IH}			L	L	H	H

1/ I_{IH} - The presence of a differential input voltage greater than the minimum input threshold voltage and providing the minimum high state current will cause a change in output logic state.

2/ I_{IL} - The presence of a differential input voltage less than the minimum input threshold voltage or providing less than the maximum input low-state current will not cause a change in the output logic state.



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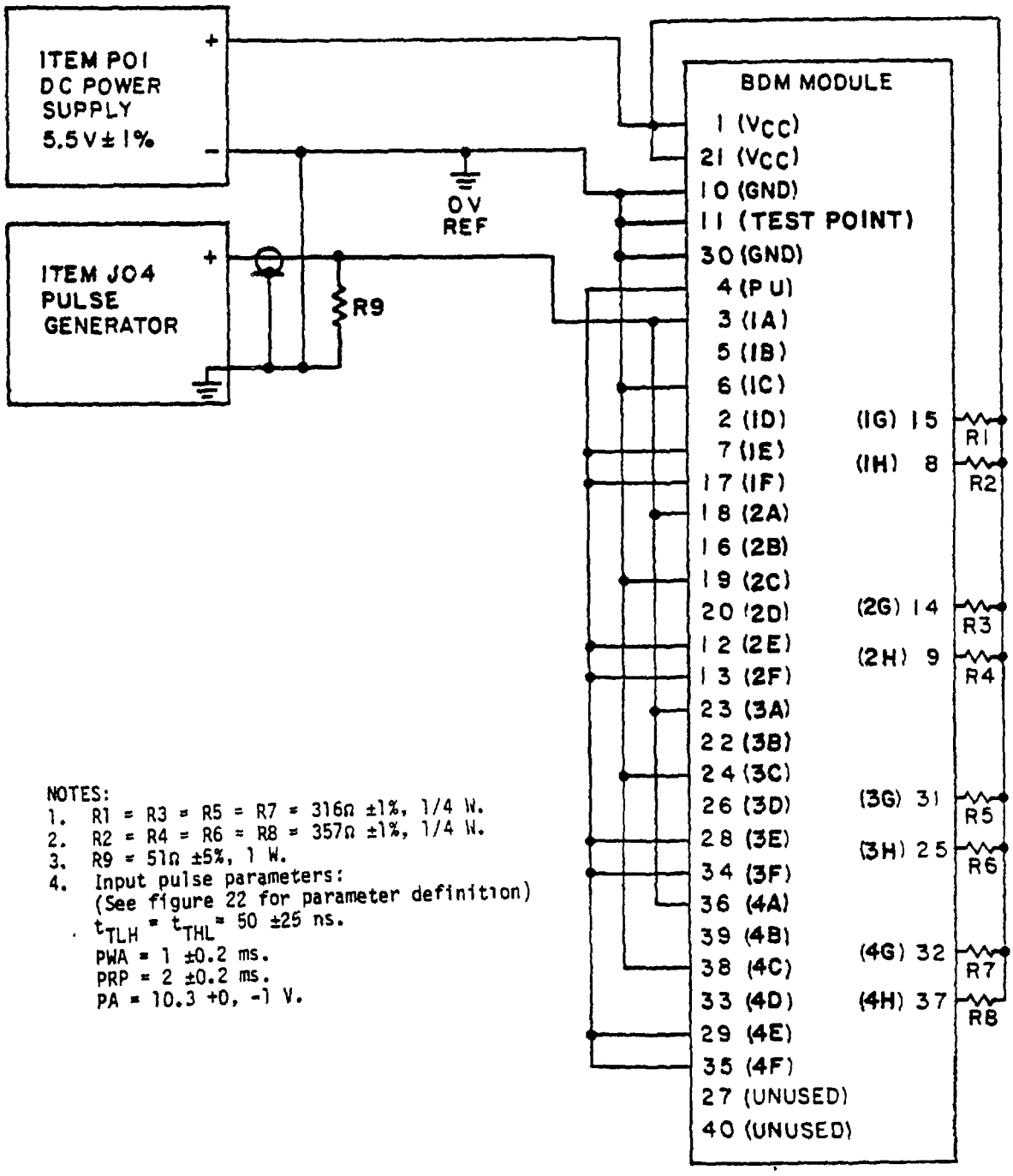
6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5963-N425)

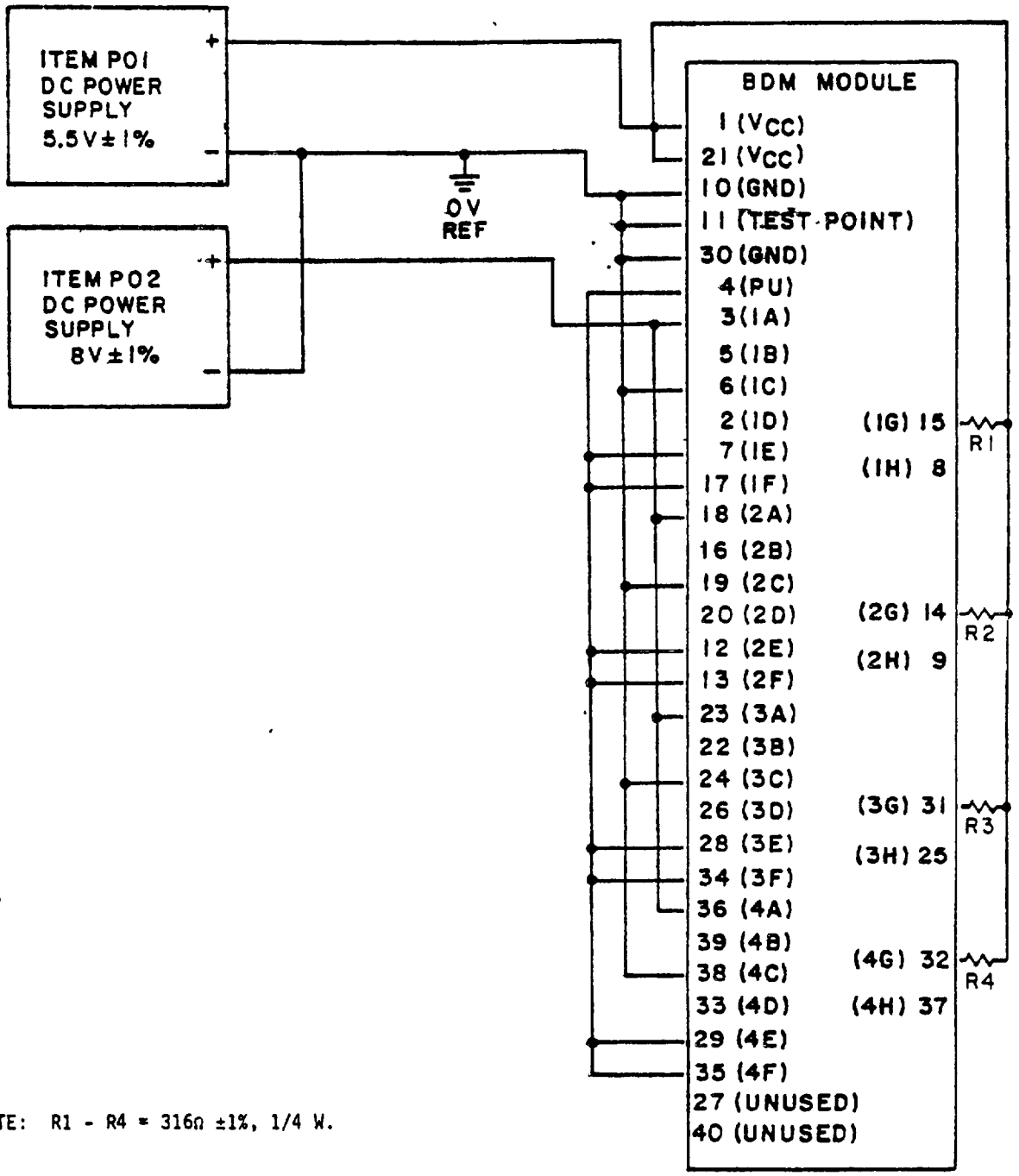
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NOTES:
1. R1 = R3 = R5 = R7 = 316Ω ±1%, 1/4 W.
2. R2 = R4 = R6 = R8 = 357Ω ±1%, 1/4 W.
3. R9 = 51Ω ±5%, 1 W.
4. Input pulse parameters:
   (See figure 22 for parameter definition)
   $t_{TLH} = t_{THL} = 50 \pm 25$ ns.
   PWA = 1 ±0.2 ms.
   PRP = 2 ±0.2 ms.
   PA = 10.3 +0, -1 V.

* FIGURE 2. Operating temperature and life test circuit.

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NOTE: R1 - R4 = 316Ω ±1%, 1/4 W.

FIGURE 3. Component temperature test circuit.

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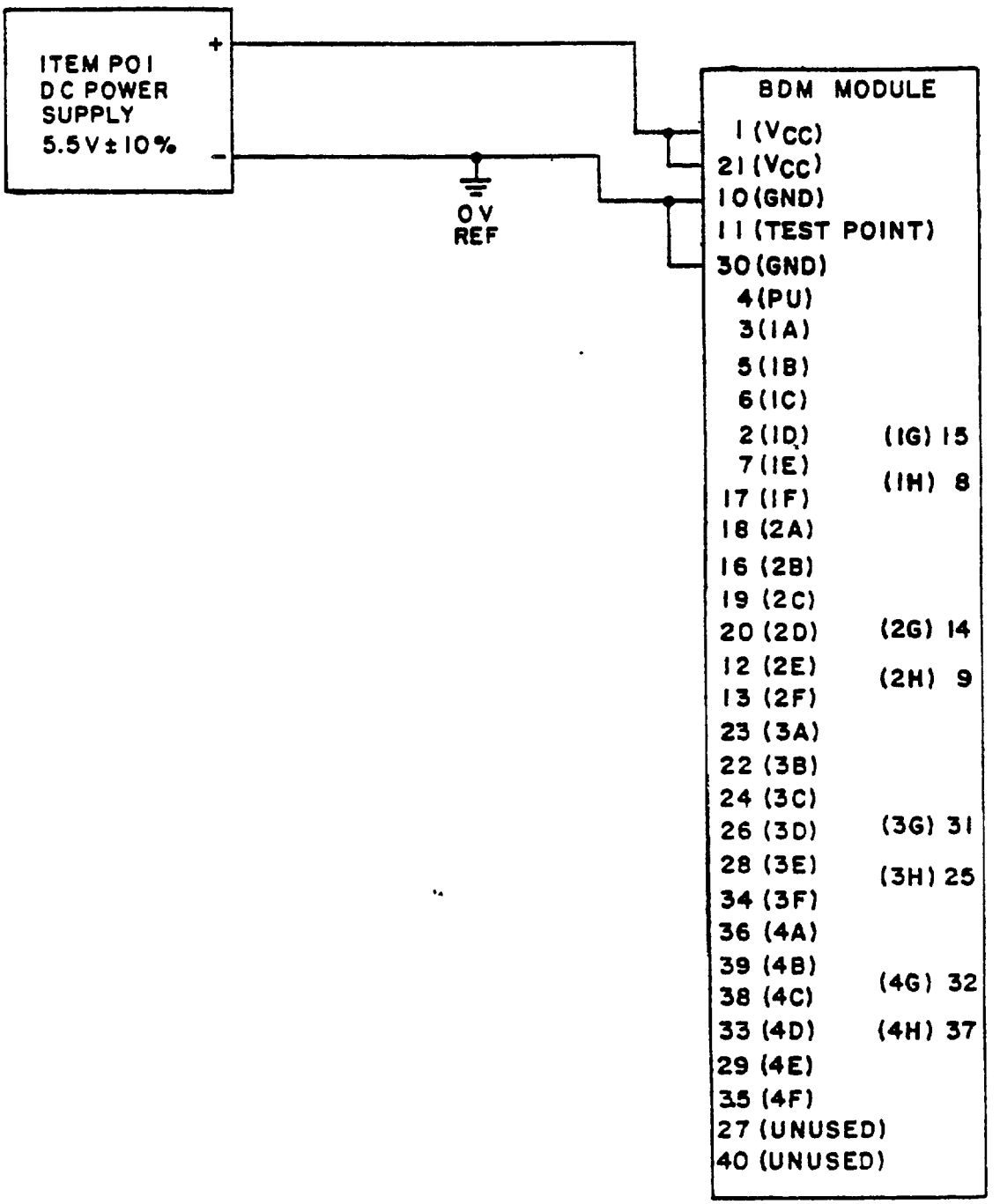


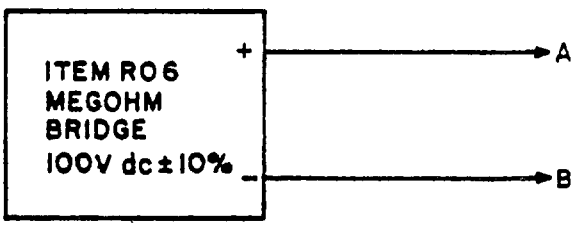
FIGURE 4. Vibration test circuit.

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TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation (pin 11)	1 thru 10, 12 thru 40	11	10	∞	MΩ	5
A2	Isolation (unused pins)	1 thru 26, 28 thru 39	27,40	"	"	"	"
A3	Isolation (used pins)	2,3,5,6,40	1,4,7 thru 39	"	"	"	"
A4	"	16,18,19,20	1 thru 15,17, 21 thru 40	"	"	"	"
A5	"	22,23,24,26	1 thru 21,25, 27 thru 40	"	"	"	"
A6	"	33,36,38,39	1 thru 32,34, 35,37,40	"	"	"	"

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the megohm bridge at "discharge" before connecting or disconnecting the test leads.

BDM MODULE	
1 (V_{CC})	
21 (V_{CC})	
10 (GND)	
11 (TEST POINT)	
30 (GND)	
4 (PU)	
3 (1A)	
5 (1B)	
6 (1C)	
2 (1D)	(1G) 15
7 (1E)	(1H) 8
17 (1F)	
18 (2A)	
16 (2B)	
19 (2C)	
20 (2D)	(2G) 14
12 (2E)	(2H) 9
13 (2F)	
23 (3A)	
22 (3B)	
24 (3C)	
26 (3D)	(3G) 31
28 (3E)	(3H) 25
34 (3F)	
36 (4A)	
39 (4B)	(4G) 32
38 (4C)	
33 (4D)	(4H) 37
29 (4E)	
35 (4F)	
27 (UNUSED)	
40 (UNUSED)	

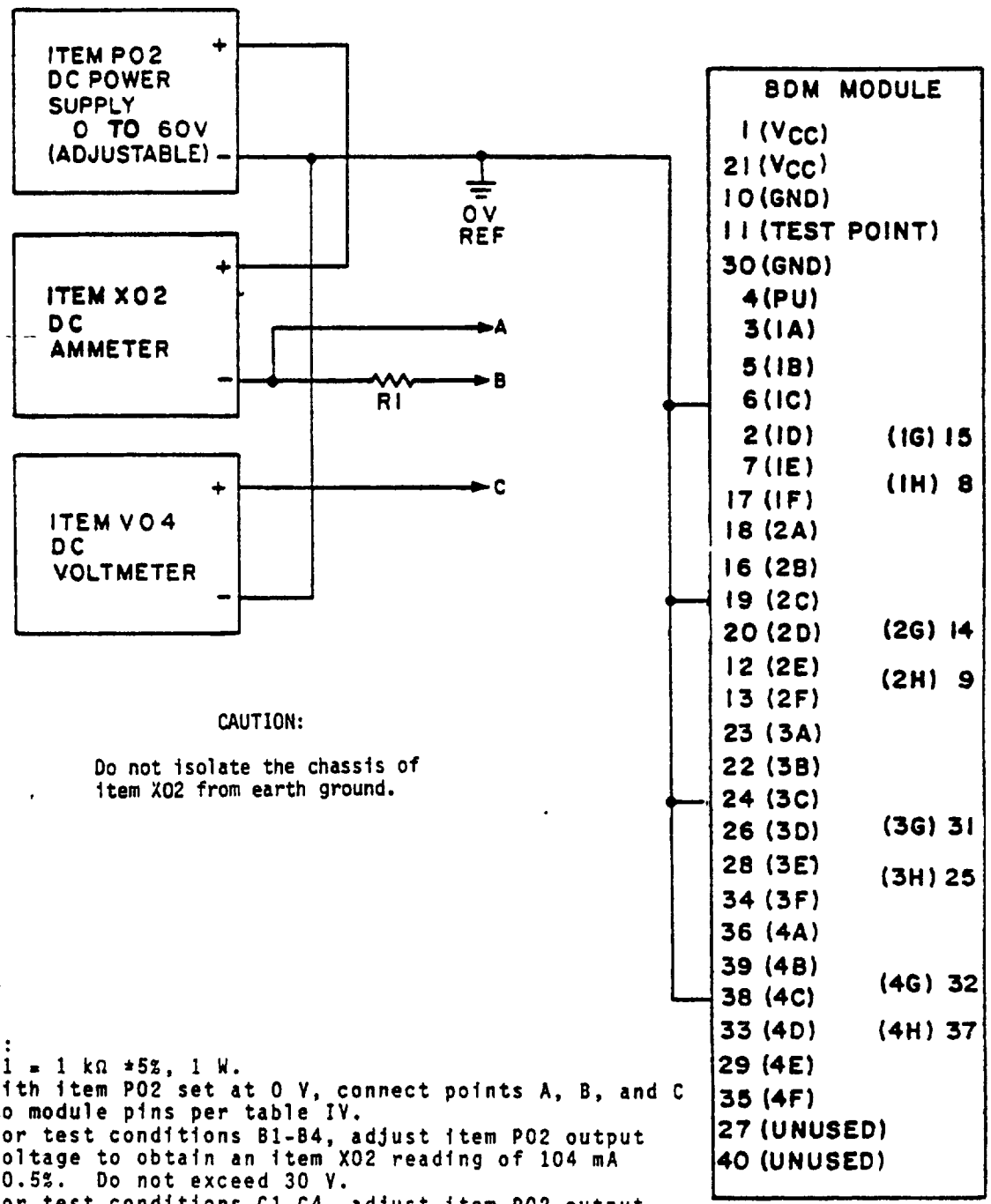
FIGURE 5 Isolation test circuit for table III.

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* TABLE IV. Termination resistor voltage and LED forward voltage (high-level and low-level) test.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
B1	1D	Termination resistor voltage	2	Open	2	9.31	9.60	V	6
B2	2D	"	20	"	20	"	"	"	"
B3	3D	"	26	"	26	"	"	"	"
B4	4D	"	33	"	33	"	"	"	"
C1	1B	LED forward voltage, high level	Open	5	5	2.20	2.45	"	"
C2	2B	"	"	16	16	"	"	"	"
C3	3B	"	"	22	22	"	"	"	"
C4	4B	"	"	39	39	"	"	"	"
C5	1B	LED forward voltage, low level	"	5	5	1.3	2.3	"	"
C6	2B	"	"	16	16	"	"	"	"
C7	3B	"	"	22	22	"	"	"	"
C8	4B	"	"	39	39	"	"	"	"

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CAUTION:

Do not isolate the chassis of item X02 from earth ground.

NOTES:
1. R1 = 1 kΩ ±5%, 1 W.
2. With item P02 set at 0 V, connect points A, B, and C to module pins per table IV.
3. For test conditions B1-B4, adjust item P02 output voltage to obtain an item X02 reading of 104 mA ±0.5%. Do not exceed 30 V.
4. For test conditions C1-C4, adjust item P02 output voltage to obtain an item X02 reading of 20 mA ±0.5%. Do not exceed 30 V.
5. For test conditions C5-C8, adjust item P02 output voltage to obtain an item X02 reading of 0.25 mA ±0.5%. Do not exceed 30 V.

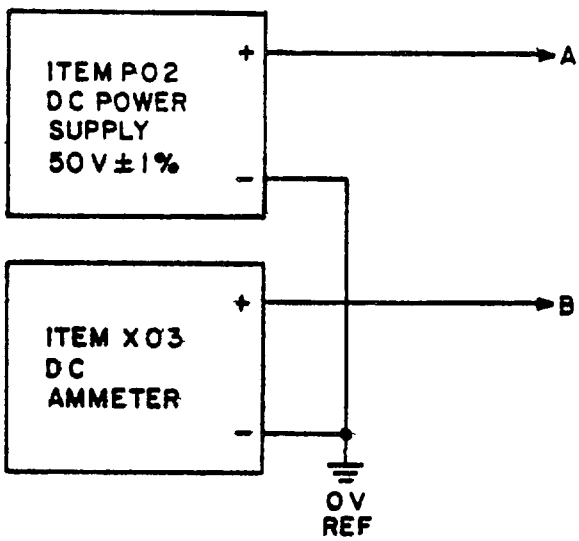
* FIGURE 6. Termination resistor voltage and LED forward voltage (high-level and low-level) test circuit for table IV.

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TABLE V. LED reverse breakdown voltage (measuring leakage current) test.

Test condition		Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
D1	1B	LED reverse breakdown voltage (measuring leakage current)	6	3	0.0	0.5	µA	7
D2	2B	"	19	18	"	"	"	"
D3	3B	"	24	23	"	"	"	"
D4	4B	"	38	36	"	"	"	"

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CAUTION:

Do not isolate the chassis of item X03 from earth ground.

BDM MODULE	
1 (V_{CC})	
21 (V_{CC})	
10 (GND)	
11 (TEST POINT)	
30 (GND)	
4 (PU)	
3 (1A)	
5 (1B)	
6 (1C)	
2 (1D)	(1G) 15
7 (1E)	(1H) 8
17 (1F)	
18 (2A)	
16 (2B)	
19 (2C)	
20 (2D)	(2G) 14
12 (2E)	(2H) 9
13 (2F)	
23 (3A)	
22 (3B)	
24 (3C)	
26 (3D)	(3G) 31
28 (3E)	(3H) 25
34 (3F)	
36 (4A)	
39 (4B)	(4G) 32
38 (4C)	
33 (4D)	(4H) 37
29 (4E)	
35 (4F)	
27 (UNUSED)	
40 (UNUSED)	

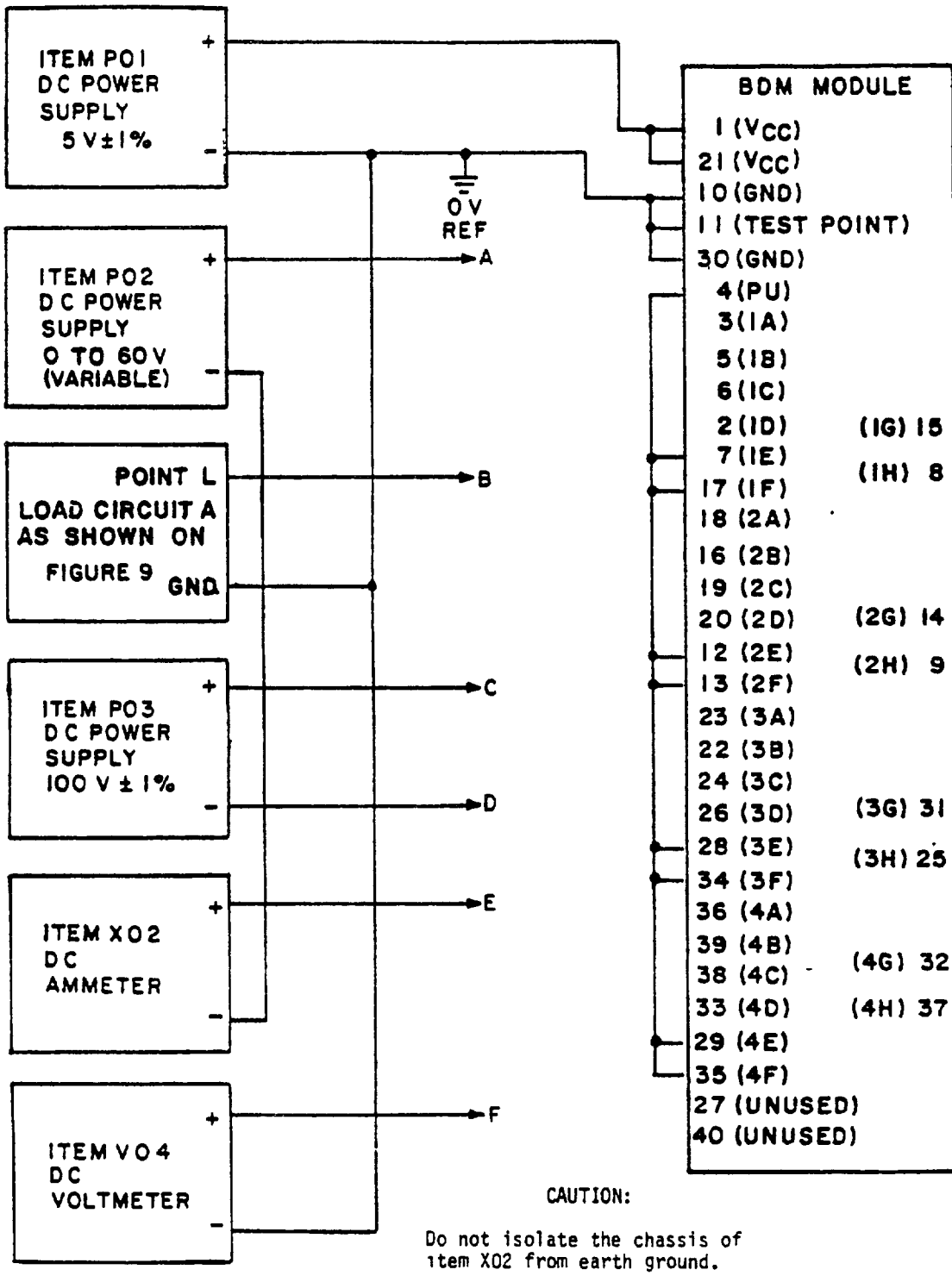
FIGURE 7. LED reverse breakdown voltage (measuring leakage current) test circuit for table V.

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* TABLE VI. Common mode voltage (low-level and high-level) test.

Test condition	Parameter		Connect						Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
E1	1G	Common mode voltage, low level	3	15	6	10,30	6	15	20	400	mV	8
E2	2G	"	18	14	19	"	19	14	"	"	"	"
E3	3G	"	23	31	24	"	24	31	"	"	"	"
E4	4G	"	36	32	38	"	38	32	"	"	"	"
E5	1G	"	3	15	10,30	6	6	15	"	"	"	"
E6	2G	"	18	14	"	19	19	14	"	"	"	"
E7	3G	"	23	31	"	24	24	31	"	"	"	"
E8	4G	"	36	32	"	38	38	32	"	"	"	"
E9	1G	Common mode voltage, high level	3	15	6	10,30	6	15	2.4	5.0	V	"
E10	2G	"	18	14	19	"	19	14	"	"	"	"
E11	3G	"	23	31	24	"	24	31	"	"	"	"
E12	4G	"	36	32	38	"	38	32	"	"	"	"
E13	1G	"	3	15	10,30	6	6	15	"	"	"	"
E14	2G	"	18	14	"	19	19	14	"	"	"	"
E15	3G	"	23	31	"	24	24	31	"	"	"	"
E16	4G	"	36	32	"	38	38	32	"	"	"	"

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CAUTION:

Do not isolate the chassis of item X02 from earth ground.

WARNING:

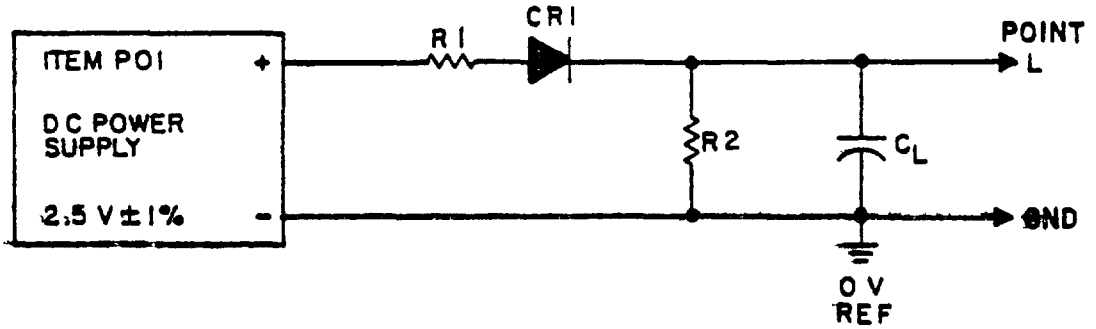
The ground return of items P01 and P03 shall be electrically isolated from the ground return of item P02.

NOTES:
1. Adjust item P02 for 8 mA ±0.5% indication on item X02 for test conditions E1-E8.
2. Adjust item P02 for 0.25 mA ±0.5% indication on item X02 for test conditions E9-E16.

* FIGURE 8. Common mode voltage (low-level and high-level) test circuit for table VI.

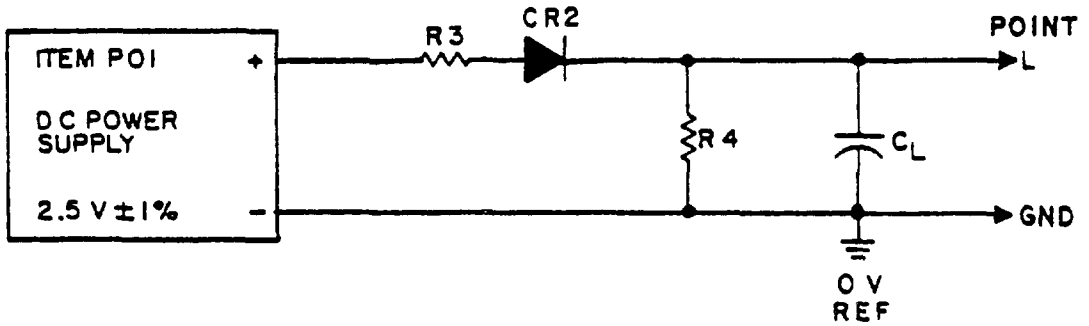
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Load A (load for output pins 15, 14, 31, and 32).



NOTE:
1. CR1 = 1N4150 or equivalent.
2. R1 = Select value for a current flow of 16 mA ±5% out of point L with point L held at 0.4 V ±1%.
3. R2 = 6.04 kΩ ±1%, 1/2 W.
4. C_L = 50±5 pF (including probe and parasitic capacitance) for test conditions E1-E16, P5-P8, Q5-Q8, U5-U8, V5-V8, W5-W8, X5-X8, Y5-Y12, Z5-Z12, CC1-CC16, and DD1-DD16.
   = 150 ±5 pF (including probe and parasitic capacitance) for test conditions U13-U16, V13-V16, W13-W16, Y17-Y24, and Z17-Z24.
5. C_L is not required for dc electrical tests E1-E16.

Load B (load for output pins 8, 9, 25, and 37).



NOTES:
1. CR2 = 1N4150 or equivalent.
2. R3 = Select value for a current flow of 14.4 mA ±5% out of point L with point L at 0.4 V ±1%.
3. R4 = 6.65 kΩ ±1%, 1/8 W.
4. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions M1-M4, N1-N4, P1-P4, Q1-Q4, U1-U8, V1-V8, W1-W8, X1-X8, Y1-Y8, Z1-Z8, AA1-AA4, BB1-BB4, CC1-CC16, and DD1-DD16.
   C_L = 150 ±5 pF (including probe and parasitic capacitance) for test conditions U9-U16, V9-V16, W9-W16, X9-X16, Y13-Y20, and Z13-Z20.
5. C_L is not required for dc electrical tests M1-M4, N1-N4, P1-P4, and Q1-Q4.

* FIGURE 9. BDM logic load circuits.

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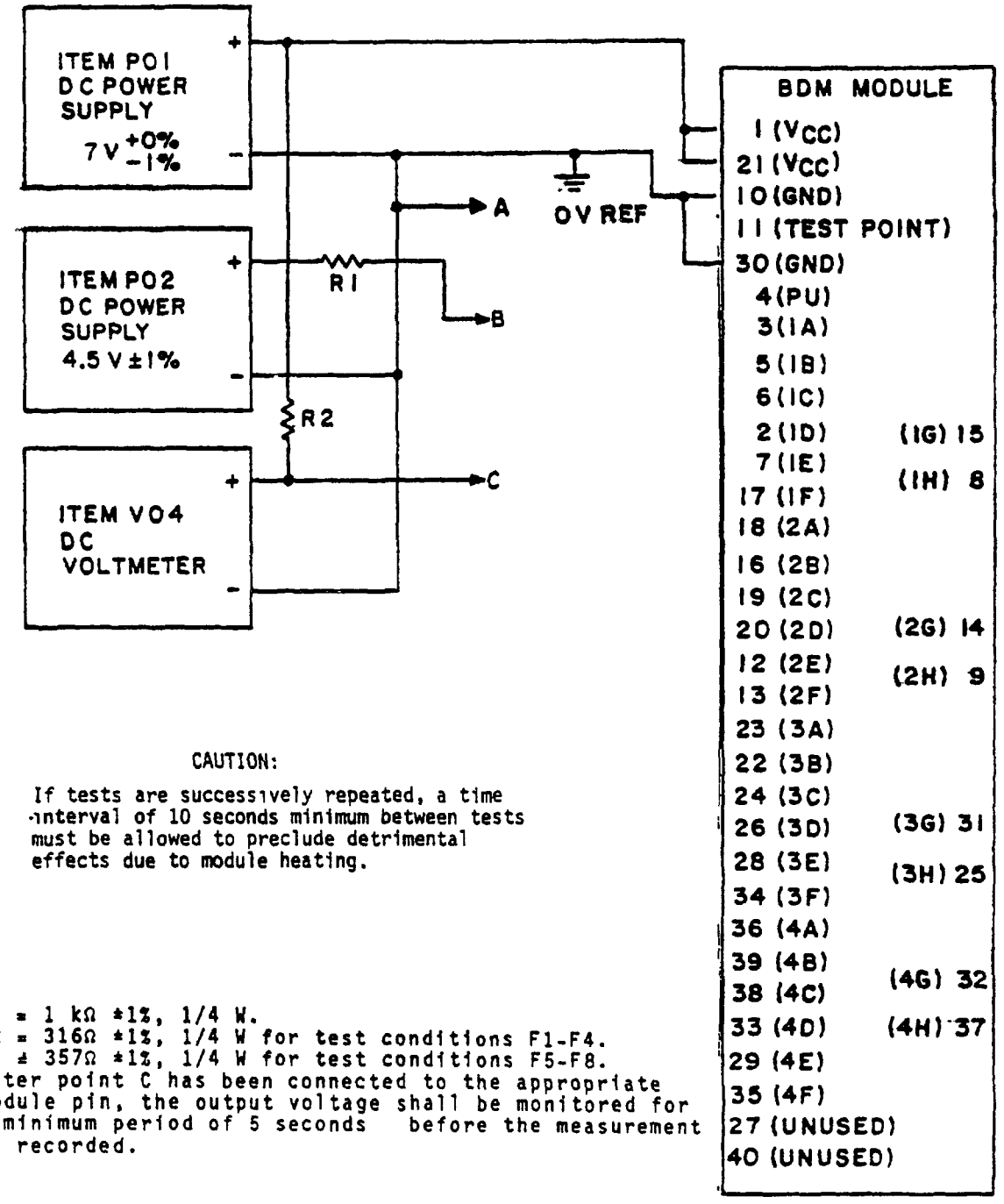
* TABLE VII. V_{OUT} (low-loaded) test.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
F1	1G	V_{OUT} (low loaded)	7,12,28,29	17,13,34, 35	15	50	500	mV	10
F2	2G	"	"	"	14	"	"	"	"
F3	3G	"	"	"	31	"	"	"	"
F4	4G	"	"	"	32	"	"	"	"
F5	1H	"	Open	7,12,28,29	8	"	"	"	"
F6	2H	"	"	"	9	"	"	"	"
F7	3H	"	"	"	25	"	"	"	"
F8	4H	"	"	"	37	"	"	"	"

* TABLE VIII. Input breakdown current test.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
G1	1E	Input breakdown current	Open	3,18,23, 36	7	0	1	mA	11
G2	2E	"	"	"	12	"	"	"	"
G3	3E	"	"	"	28	"	"	"	"
G4	4E	"	"	"	29	"	"	"	"
G5	1F	"	7,12,28,29	Open	17	"	"	"	"
G6	2F	"	"	"	13	"	"	"	"
G7	3F	"	"	"	34	"	"	"	"
G8	4F	"	"	"	35	"	"	"	"

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CAUTION:

If tests are successively repeated, a time interval of 10 seconds minimum between tests must be allowed to preclude detrimental effects due to module heating.

NOTES:
1. R1 = 1 kΩ ±1%, 1/4 W.
2. R2 = 316Ω ±1%, 1/4 W for test conditions F1-F4.
   = 357Ω ±1%, 1/4 W for test conditions F5-F8.
3. After point C has been connected to the appropriate module pin, the output voltage shall be monitored for a minimum period of 5 seconds before the measurement is recorded.

FIGURE 10. V_{OUT} (low-loaded) test circuit for table VII.

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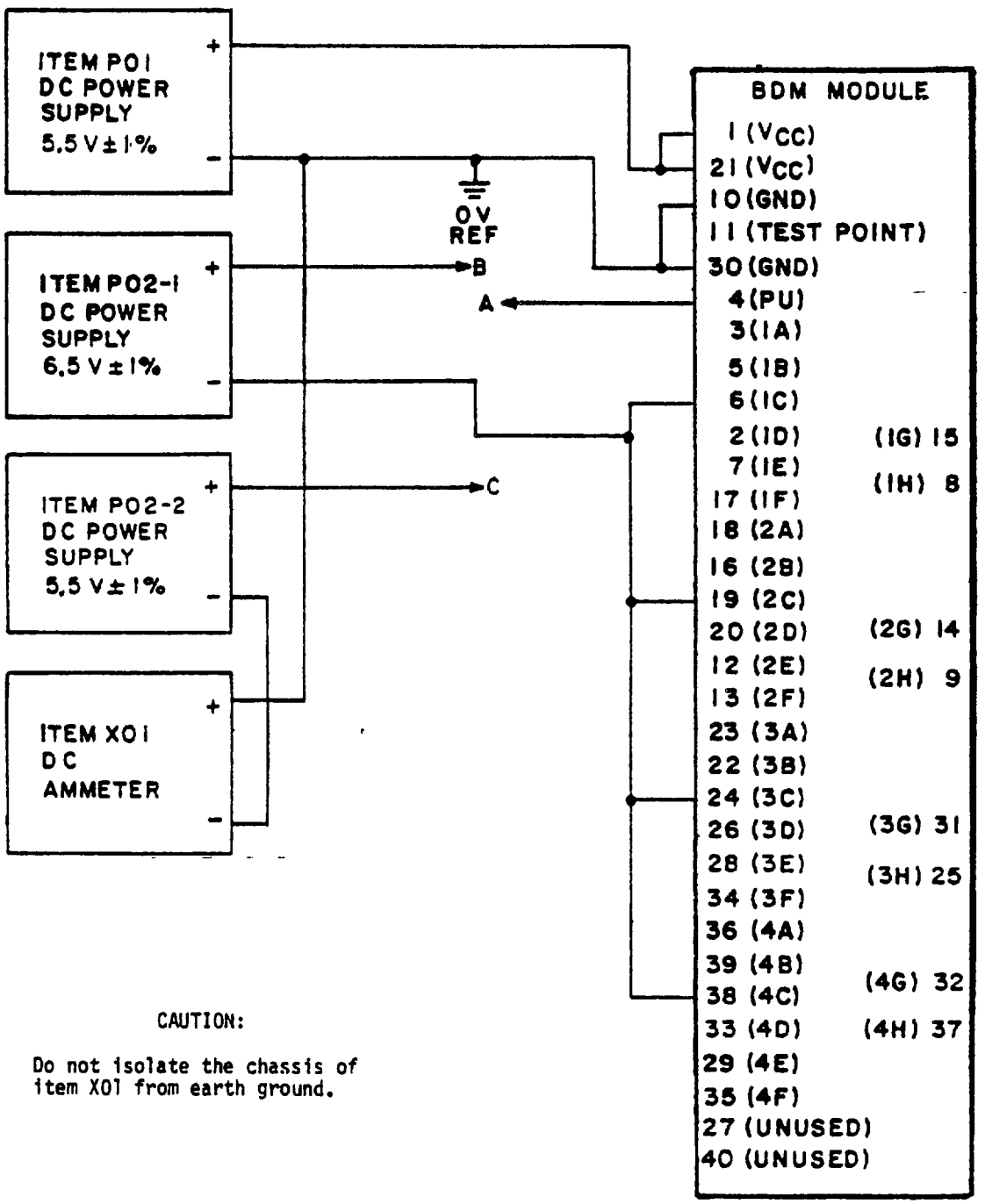


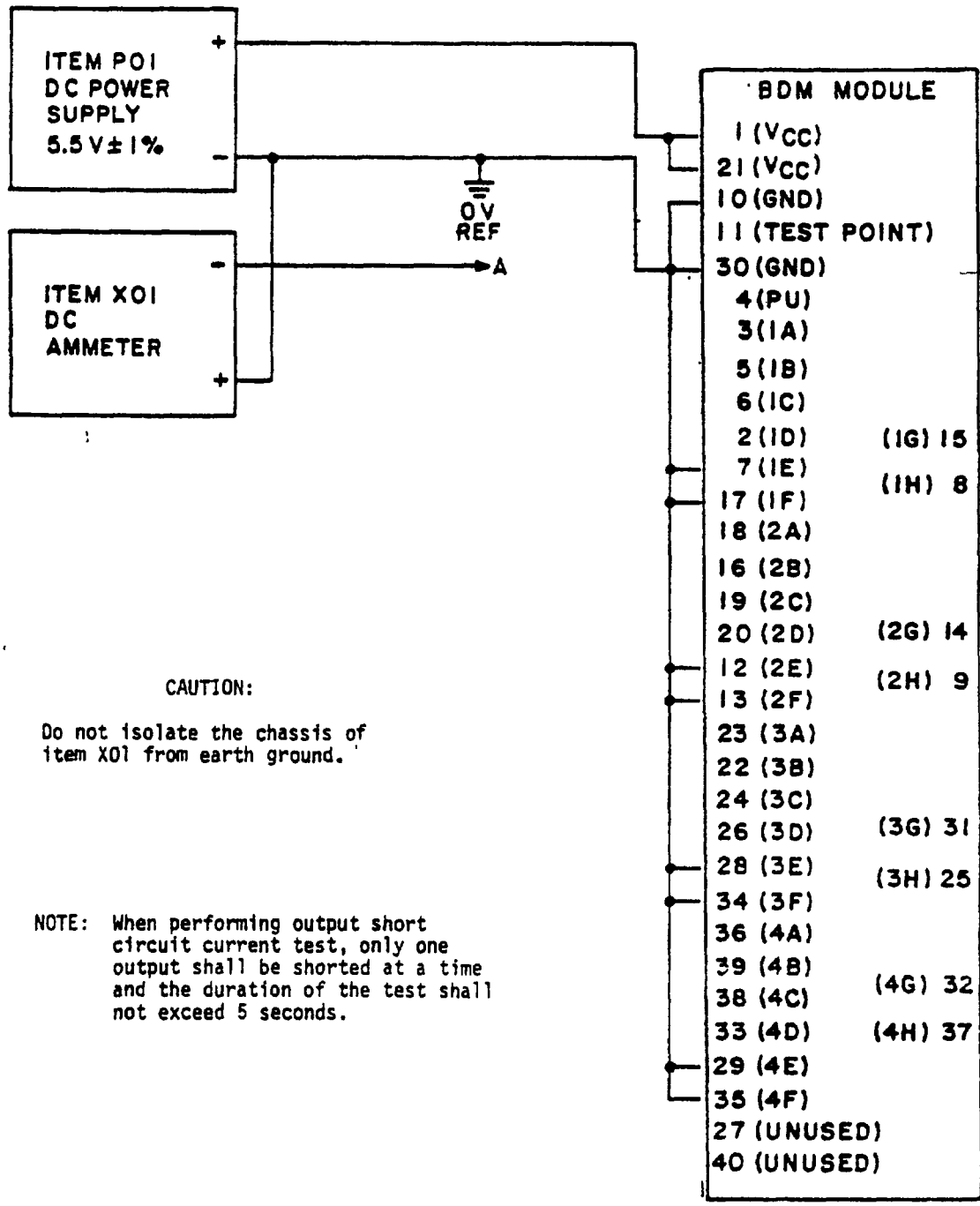
FIGURE 11. Input breakdown current test circuit for table VIII.

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TABLE IX. Output short circuit current test.

Test condition	Parameter		Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
H1	1G	Output short circuit current	15	-20	-55	mA	12
H2	2G	"	14	"	"	"	"
H3	3G	"	31	"	"	"	"
H4	4G	"	32	"	"	"	"
H5	1H	"	8	-20.0	-56.7	"	"
H6	2H	"	9	"	"	"	"
H7	3H	"	25	"	"	"	"
H8	4H	"	37	"	"	"	"

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CAUTION:

Do not isolate the chassis of item X01 from earth ground.

NOTE: When performing output short circuit current test, only one output shall be shorted at a time and the duration of the test shall not exceed 5 seconds.

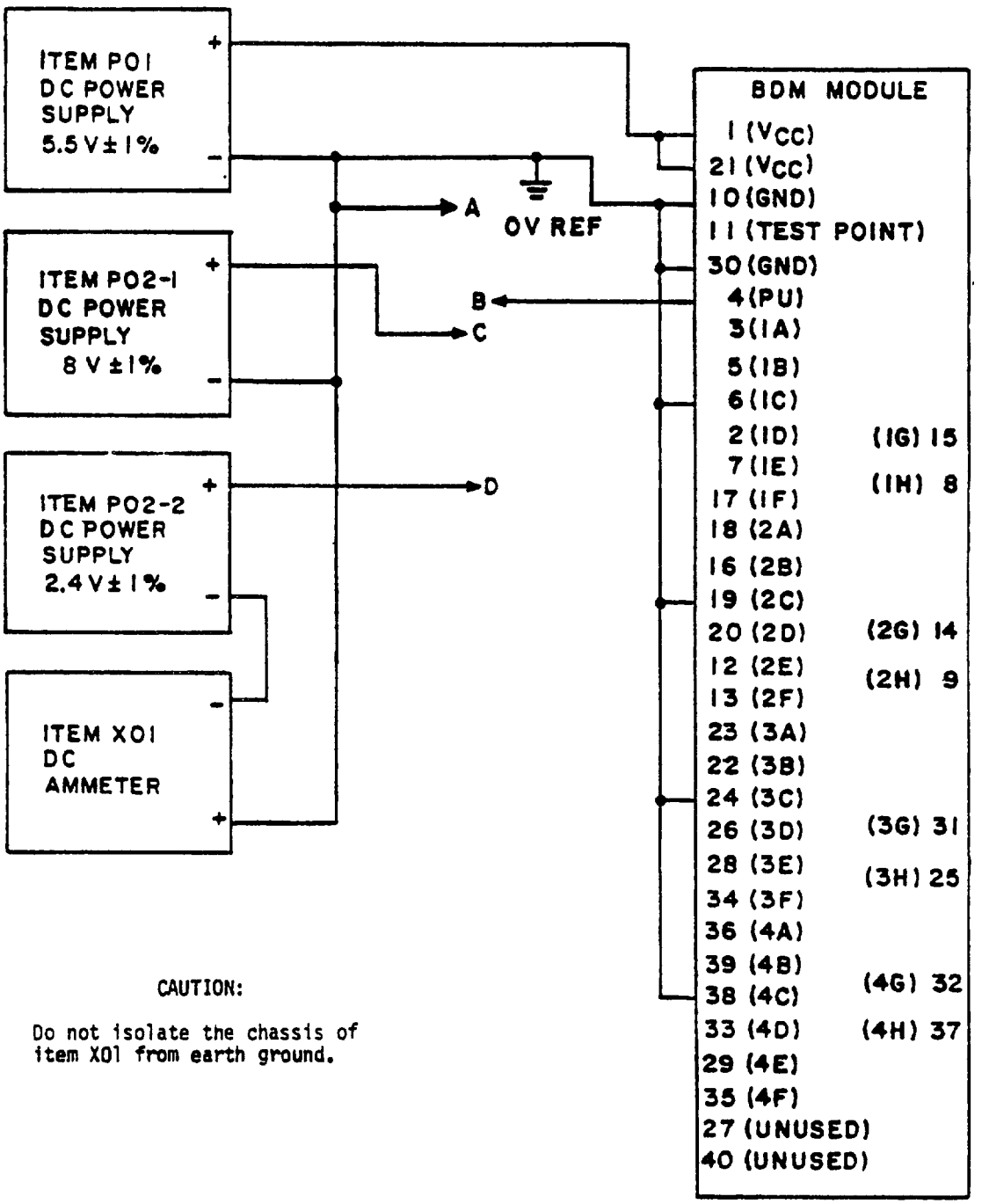
FIGURE 12. Output short circuit current test circuit for table IX.

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TABLE X. Input current (high-level) test.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
J1	1F	Input current, high level	Open	7,12, 28,29	Open	17	0	40	µA	13
J2	2F	"	"	"	"	13	"	"	"	"
J3	3F	"	"	"	"	34	"	"	"	"
J4	4F	"	"	"	"	35	"	"	"	"
J5	1E	"	12,28, 29	Open	3,18, 23,36	7	"	"	"	"
J6	2E	"	7,28, 29	"	"	12	"	"	"	"
J7	3E	"	7,12, 29	"	"	28	"	"	"	"
J8	4E	"	7,12, 28	"	"	29	"	"	"	"

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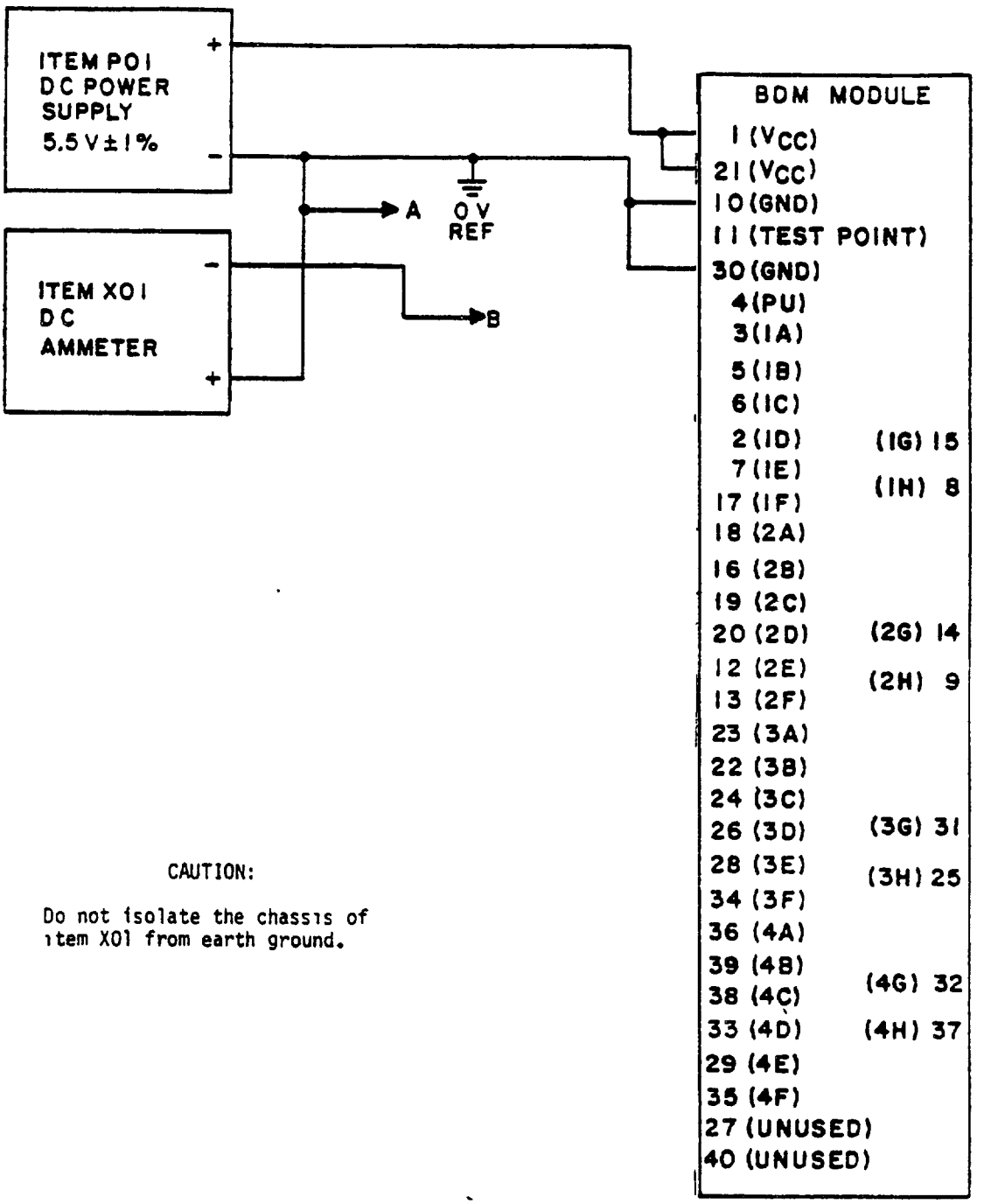
* FIGURE 13. Input current (high-level) test circuit for table X.

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TABLE XI. Input current (low-level) test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
K1	1F	Input current, low level	7,12,28,29	17	-0.70	-1.73	mA	14
K2	2F	"	"	13	"	"	"	"
K3	3F	"	"	34	"	"	"	"
K4	4F	"	"	35	"	"	"	"
K5	1E	"	Open	7	"	"	"	"
K6	2E	"	"	12	"	"	"	"
K7	3E	"	"	28	"	"	"	"
K8	4E	"	"	29	"	"	"	"

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CAUTION:

Do not isolate the chassis of item X01 from earth ground.

* FIGURE 14. Input current (low-level) test circuit for table XI.

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* TABLE XII. Functional test.

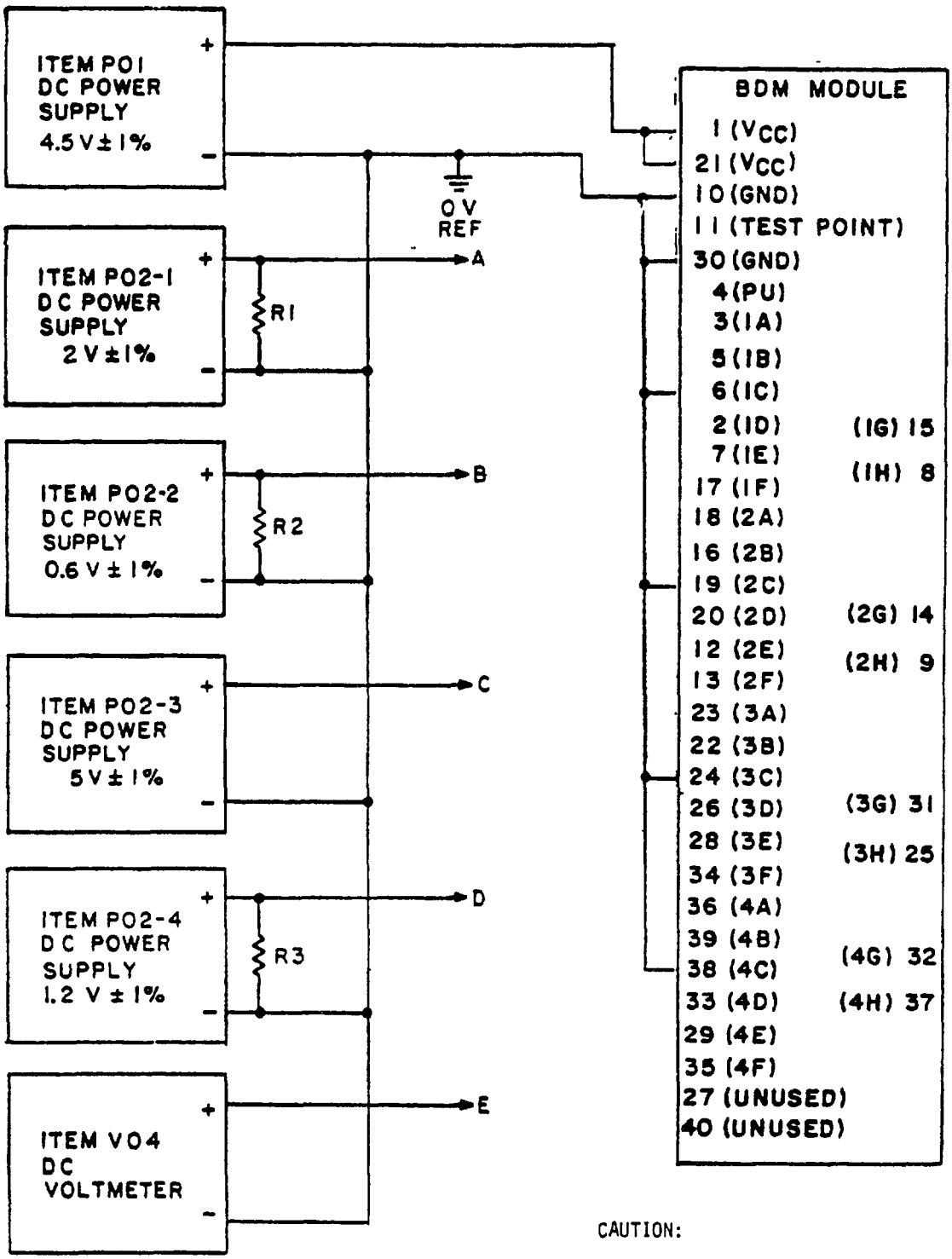
Test condition	Parameter	Connect				Output logic states 2/ 3/								Initial limits 25°C ±5°C ambient		Test figure
		A to pin	B to pin	C to pin	D to pin	1H	1G	2H	2G	3H	3G	4H	4G	Min	Max	
L1	Functional test	Open	7,12,13,17, 28,29,34,35	Open	3,18,23,36	H	H	H	H	H	H	H	H	1/		15
L2	"	7,12,28,29	13,17,34,35	"	"	L	H	L	H	L	H	L	H	"		"
L3	"	7,13,17,28, 34,35	12,29	"	"	L	H	H	L	L	H	H	L	"		"
L4	"	12,13,17,29, 34,35	7,28	"	"	H	L	L	H	H	L	L	H	"		"
L5	"	Open	7,12,13,17, 28,29,34,35	3,18,23,36	Open	H	H	H	H	H	H	H	H	"		"
L6	"	13,17,34,35	7,12,28,29	"	"	H	L	H	L	H	L	H	L	"		"
L7	"	7,12,28,29	13,17,34,35	"	"	H	H	H	H	H	H	H	H	"		"
L8	"	7,12,13,17, 28,29,34,35	Open	"	"	H	L	H	L	H	L	H	L	"		"

1/ Output logic states:
L = Low level = 0.0 to 0.4 V.
H = High level = 2.4 to 4.5 V.

2/ For test conditions L1-L8, verification of the output logic states in footnote 1 is obtained by connecting item V04 (point E) to each output pin in succession after input conditions have been accomplished.

3/ 1H, 1G, 2H, 2G, 3H, 3G, 4H, 4G output logic states are measured at output pins 8, 15, 9, 14, 25, 31, 37, and 32 respectively.

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CAUTION:

Do not isolate the chassis of item X01 from earth ground.

NOTE: R1 = R2 = R3 = 51Ω ±5%, 1 W.

FIGURE 15. Functional test circuit for table XII.

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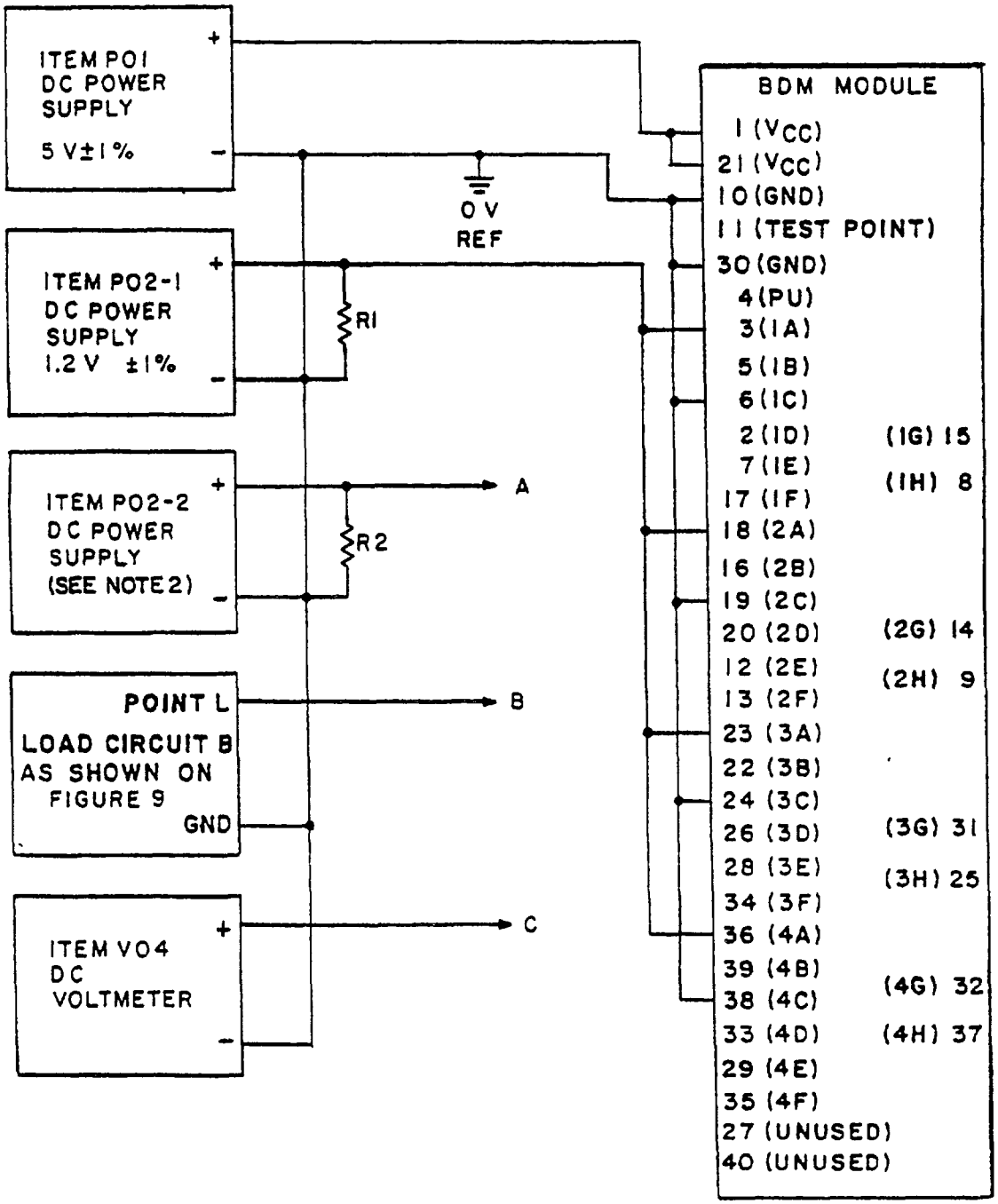
* TABLE XIII. Input threshold voltage (high to low) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
M1	Input threshold voltage (high to low)	7	8	8	0.8	2.0	V	16
M2	"	12	9	9	"	"	"	"
M3	"	28	25	25	"	"	"	"
M4	"	29	37	37	"	"	"	"

* TABLE XIV. Input threshold voltage (low to high) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
N1	Input threshold voltage (low to high)	7	8	8	0.6	2.0	V	16
N2	"	12	9	9	"	"	"	"
N3	"	28	25	25	"	"	"	"
N4	"	29	37	37	"	"	"	"

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* FIGURE 16. Input threshold voltage (high to low and low to high) test circuit for tables XIII and XIV.

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NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. Test procedure:
   a. Initially set item P02-2 to 0.0 V for each test condition.
   b. Verify that the output of the gate under test is in a high-level state ($V_{OUT} \geq 2.4$ V).
   c. Slowly increase the item P02-2 output voltage until the output of the gate under test switches logic states and goes to a low-level state ($V_{OUT} \leq 0.4$ V). Record the input voltage (V_{THL}) as test condition M.
   d. Continue to increase the item P02-2 output voltage until it reaches 2.0 V.
   e. Slowly decrease the item P02-2 output voltage until the output of the gate under test switches logic states and goes to a high-level state ($V_{OUT} \geq 2.4$). Record the input voltage (V_{TLH}) as test condition N.
   f. Decrease the item P02-2 output voltage to 0.0 V and repeat the test procedure for each test condition.

FIGURE 16. Input threshold voltage (high to low and low to high) test circuit for tables XIII and XIV - Continued.

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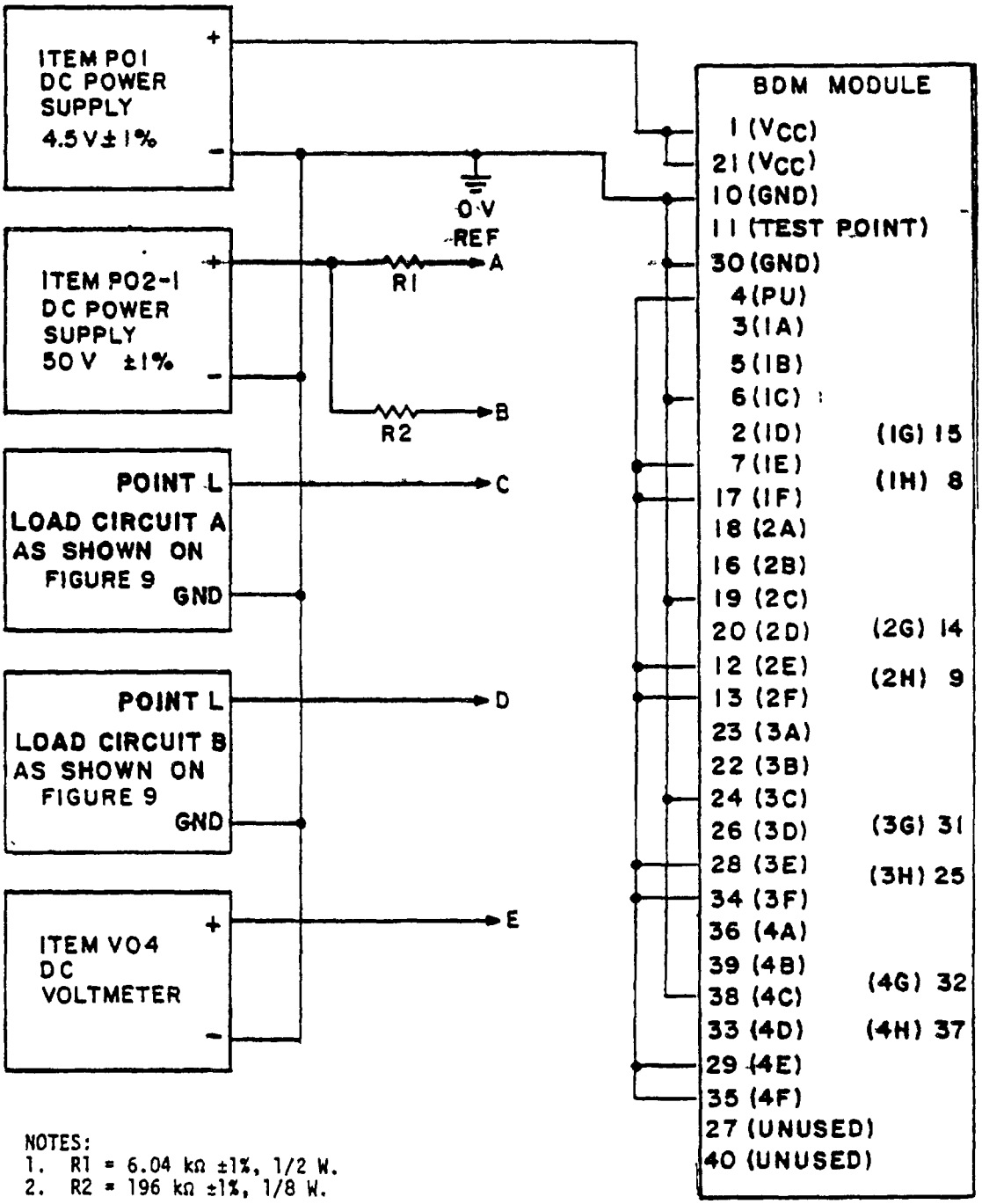
* TABLE XV. Output voltage (low-level) test.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
P1	1H Output voltage, low level	Open	5	Open	8	8	20	400	mV	17
P2	2H "	"	16	"	9	9	"	"	"	"
P3	3H "	"	22	"	25	25	"	"	"	"
P4	4H "	"	39	"	37	37	"	"	"	"
P5	1G "	5	Open	15	Open	15	"	"	"	"
P6	2G "	16	"	14	"	14	"	"	"	"
P7	3G "	22	"	31	"	31	"	"	"	"
P8	4G "	39	"	32	"	32	"	"	"	"

* TABLE XVI. Output voltage (high-level) test.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Q1	1H Output voltage, high level	5	Open	Open	8	8	2.4	4.5	V	17
Q2	2H "	16	"	"	9	9	"	"	"	"
Q3	3H "	22	"	"	25	25	"	"	"	"
Q4	4H "	39	"	"	37	37	"	"	"	"
Q5	1G "	Open	5	15	Open	15	"	"	"	"
Q6	2G "	"	16	14	"	14	"	"	"	"
Q7	3G "	"	22	31	"	31	"	"	"	"
Q8	4G "	"	39	32	"	32	"	"	"	"

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NOTES:
1. R1 = 6.04 kΩ ±1%, 1/2 W.
2. R2 = 196 kΩ ±1%, 1/8 W.

* FIGURE 17. Output voltage (low-level and high-level) test circuit for tables XV and XVI.

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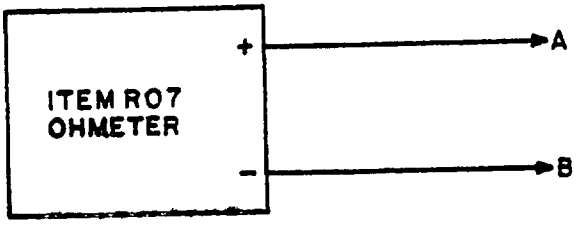
* TABLE XVII. Continuity and resistance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
R1	Continuity, power supply pin	1	21	0	1	Ω	18
R2	Continuity, ground return	10	30	0	1	"	"
R3	Resistance, current limit	3	5	293.7	306.3	"	"
R4	"	18	16	"	"	"	"
R5	"	23	22	"	"	"	"
R6	"	36	39	"	"	"	"
R7	Resistance, termination	6	2	90.0	91.8	"	"
R8	"	19	20	"	"	"	"
R9	"	24	26	"	"	"	"
R10	"	38	33	"	"	"	"
R11	Resistance, pullup	4	1	900	1,100	"	"

* TABLE XVIII. Power supply current test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
S1	Power supply current, G and H outputs high	Open	D,E	Open	20.0	78.8	mA	19
S2	Power supply current, G outputs low, H outputs high	E	Open	D	55	125	mA	19

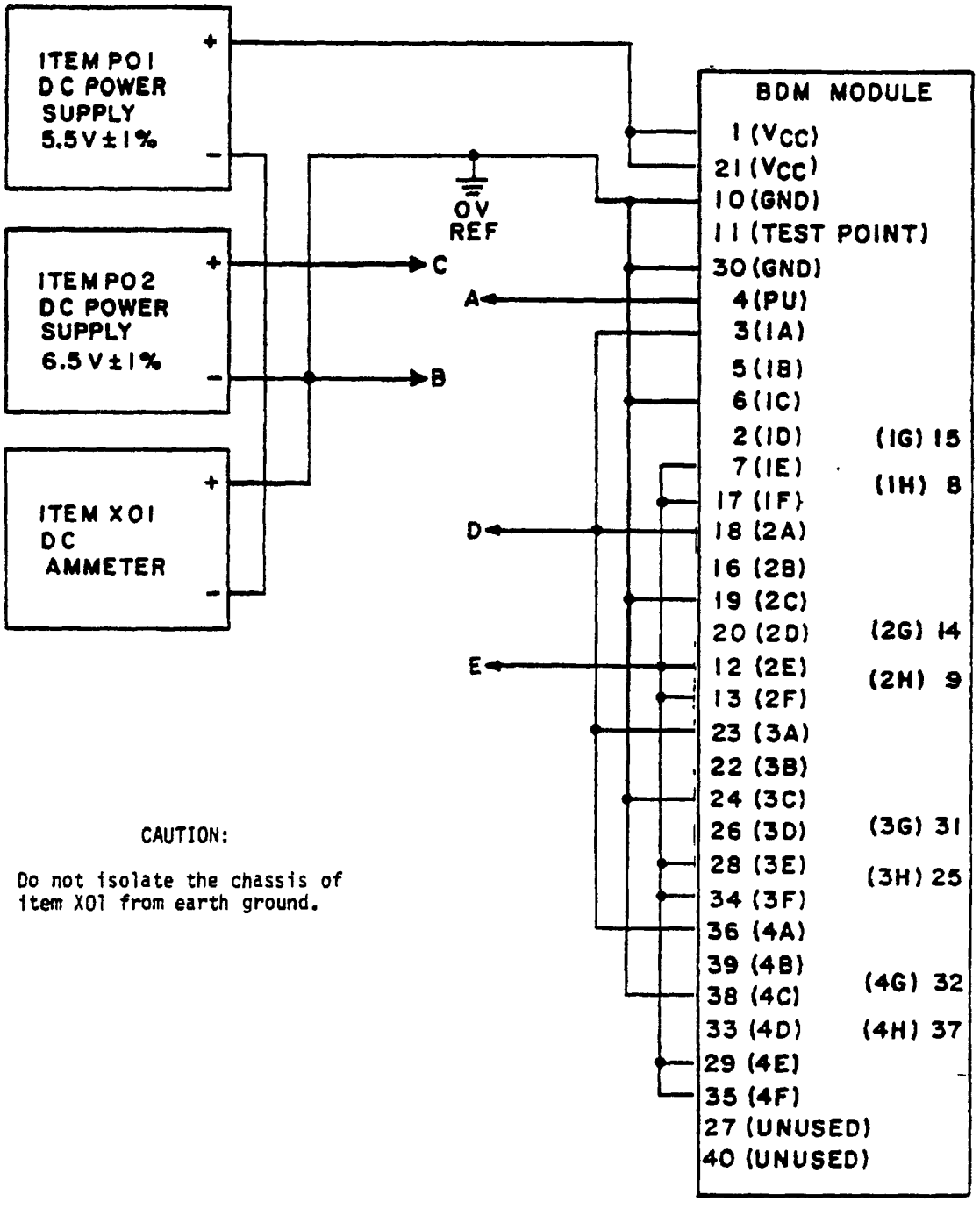
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BDM MODULE		
1 (V_{CC})		
21 (V_{CC})		
10 (GND)		
11 (TEST POINT)		
30 (GND)		
4 (PU)		
3 (1A)		
5 (1B)		
6 (1C)		
2 (1D)	(1G)	15
7 (1E)	(1H)	8
17 (1F)		
18 (2A)		
16 (2B)		
19 (2C)		
20 (2D)	(2G)	14
12 (2E)	(2H)	9
13 (2F)		
23 (3A)		
22 (3B)		
24 (3C)		
26 (3D)	(3G)	31
28 (3E)	(3H)	25
34 (3F)		
36 (4A)		
39 (4B)	(4G)	32
38 (4C)		
33 (4D)	(4H)	37
29 (4E)		
35 (4F)		
27 (UNUSED)		
40 (UNUSED)		

* FIGURE 18. Continuity and resistance test circuit for table XVII.

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CAUTION:

Do not isolate the chassis of item XO1 from earth ground.

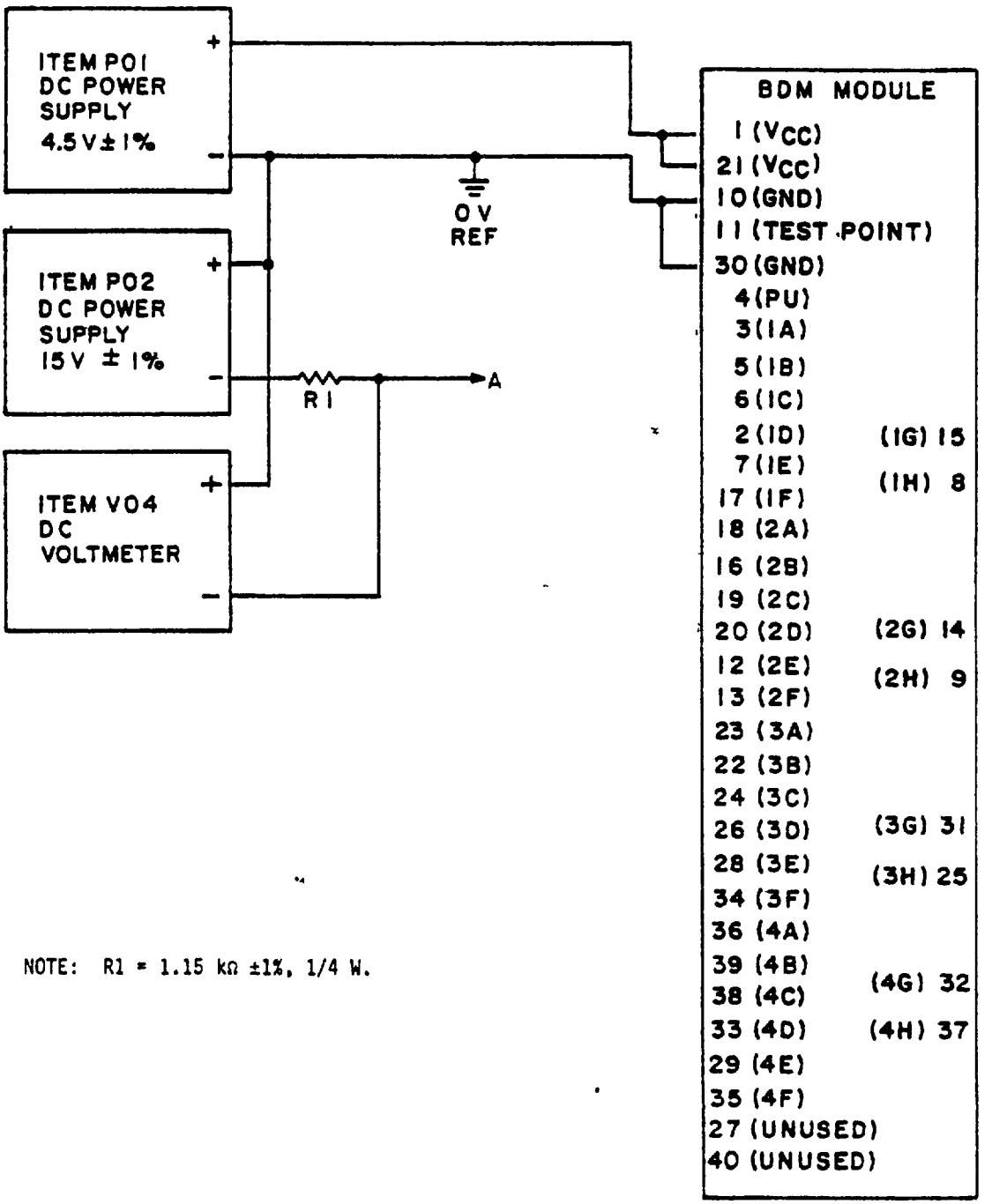
* FIGURE 19. Power supply current test circuit for table XVIII.

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* TABLE XIX. Input clamp diode voltage test.

Test condition	Parameter		Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
T1	1E	Input clamp diode voltage	7	-0.4	-1.5	V	20
T2	2E	"	12	"	"	"	"
T3	3E	"	28	"	"	"	"
T4	4E	"	29	"	"	"	"
T5	1F	"	17	"	"	"	"
T6	2F	"	13	"	"	"	"
T7	3F	"	34	"	"	"	"
T8	4F	"	35	"	"	"	"

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NOTE: R1 = 1.15 kΩ ±1%, 1/4 W.

* FIGURE 20. Input clamp diode voltage test circuit for table XIX.

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* TABLE XX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
U1	1H	Transition time (t_{TLH}), C_L = 50 pF	3	8	Open	8	1	13	ns	21,22
U2	2H	"	18	9	"	9	"	"	"	"
U3	3H	"	23	25	"	25	"	"	"	"
U4	4H	"	36	37	"	37	"	"	"	"
U5	1G	"	3	15	15	8	"	"	"	21,23
U6	2G	"	18	14	14	9	"	"	"	"
U7	3G	"	23	31	31	25	"	"	"	"
U8	4G	"	36	32	32	37	"	"	"	"
U9	1H	Transition time (t_{TLH}), C_L = 150 pF	3	8	Open	8	3	21	"	21,22
U10	2H	"	18	9	"	9	"	"	"	"
U11	3H	"	23	25	"	25	"	"	"	"
U12	4H	"	36	37	"	37	"	"	"	"
U13	1G	"	3	15	15	8	"	"	"	21,23
U14	2G	"	18	14	14	9	"	"	"	"
U15	3G	"	23	31	31	25	"	"	"	"
U16	4G	"	36	32	32	37	"	"	"	"

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* TABLE XX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
V1	1H	Transition time (t_{THL}), C_L = 50 pF	3	8	Open	8	1	10	ns	21,22
V2	2H	"	18	9	"	9	"	"	"	"
V3	3H	"	23	25	"	25	"	"	"	"
V4	4H	"	36	37	"	37	"	"	"	"
V5	1G	"	3	15	15	8	"	"	"	21,23
V6	2G	"	18	14	14	9	"	"	"	"
V7	3G	"	23	31	31	25	"	"	"	"
V8	4G	"	36	32	32	37	"	"	"	"
V9	1H	Transition time (t_{THL}), C_L = 150 pF	3	8	Open	8	"	17	"	21,22
V10	2H	"	18	9	"	9	"	"	"	"
V11	3H	"	23	25	"	25	"	"	"	"
V12	4H	"	36	37	"	37	"	"	"	"
V13	1G	"	3	15	15	8	"	"	"	21,23
V14	2G	"	18	14	14	9	"	"	"	"
V15	3G	"	23	31	31	25	"	"	"	"
V16	4G	"	36	32	32	37	"	"	"	"

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* TABLE XX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

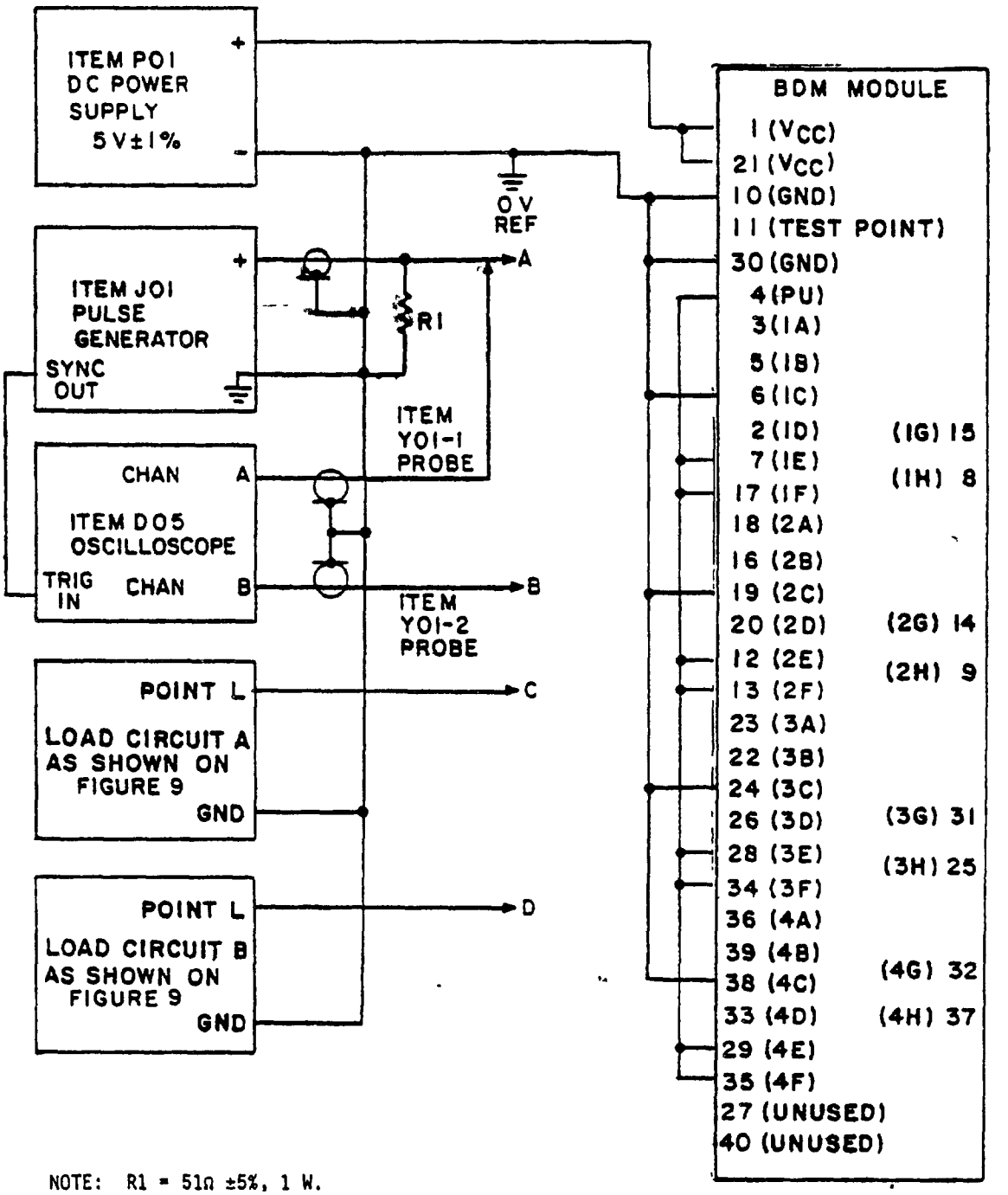
Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
W1	1A/C to 1H	Propagation delay time (t_{PLH}), C_L = 50 pF	3	8	Open	8	10	125	ns	21,22
W2	2A/C to 2H	"	18	9	"	9	"	"	"	"
W3	3A/C to 3H	"	23	25	"	25	"	"	"	"
W4	4A/C to 4H	"	36	37	"	37	"	"	"	"
W5	1A/C to 1G	"	3	15	15	8	13	145	"	21,23
W6	2A/C to 2G	"	18	14	14	9	"	"	"	"
W7	3A/C to 3G	"	23	31	31	25	"	"	"	"
W8	4A/C to 4G	"	36	32	32	37	"	"	"	"
W9	1A/C to 1H	Propagation delay time (t_{PLH}), C_L = 150 pF	3	8	Open	8	15	"	"	21,22
W10	2A/C to 2H	"	18	9	"	9	"	"	"	"
W11	3A/C to 3H	"	23	25	"	25	"	"	"	"
W12	4A/C to 4H	"	36	37	"	37	"	"	"	"
W13	1A/C to 1G	"	3	15	15	8	20	175	"	21,23
W14	2A/C to 2G	"	18	14	14	9	"	"	"	"
W15	3A/C to 3G	"	23	31	31	25	"	"	"	"
W16	4A/C to 4G	"	36	32	32	37	"	"	"	"

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* TABLE XX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
X1	1A/C to 1H Propagation delay time (t_{PHL}), C_L = 50 pF	3	8	Open	8	10	120	ns	21,22
X2	2A/C to 2H "	18	9	"	9	"	"	"	"
X3	3A/C to 3H "	23	25	"	25	"	"	"	"
X4	4A/C to 4H "	36	37	"	37	"	"	"	"
X5	1A/C to 1G "	3	15	15	8	13	145	"	21,23
X6	2A/C to 2G "	18	14	14	9	"	"	"	"
X7	3A/C to 3G "	23	31	31	25	"	"	"	"
X8	4A/C to 4G "	36	32	32	37	"	"	"	"
X9	1A/C to 1H Propagation delay time (t_{PHL}), C_L = 150 pF	3	8	Open	8	15	135	"	21,22
X10	2A/C to 2H "	18	9	"	9	"	"	"	"
X11	3A/C to 3H "	23	25	"	25	"	"	"	"
X12	4A/C to 4H "	36	37	"	37	"	"	"	"
X13	1A/C to 1G "	3	15	15	8	20	177	"	21,23
X14	2A/C to 2G "	18	14	14	9	"	"	"	"
X15	3A/C to 3G "	23	31	31	25	"	"	"	"
X16	4A/C to 4G "	36	32	32	37	"	"	"	"

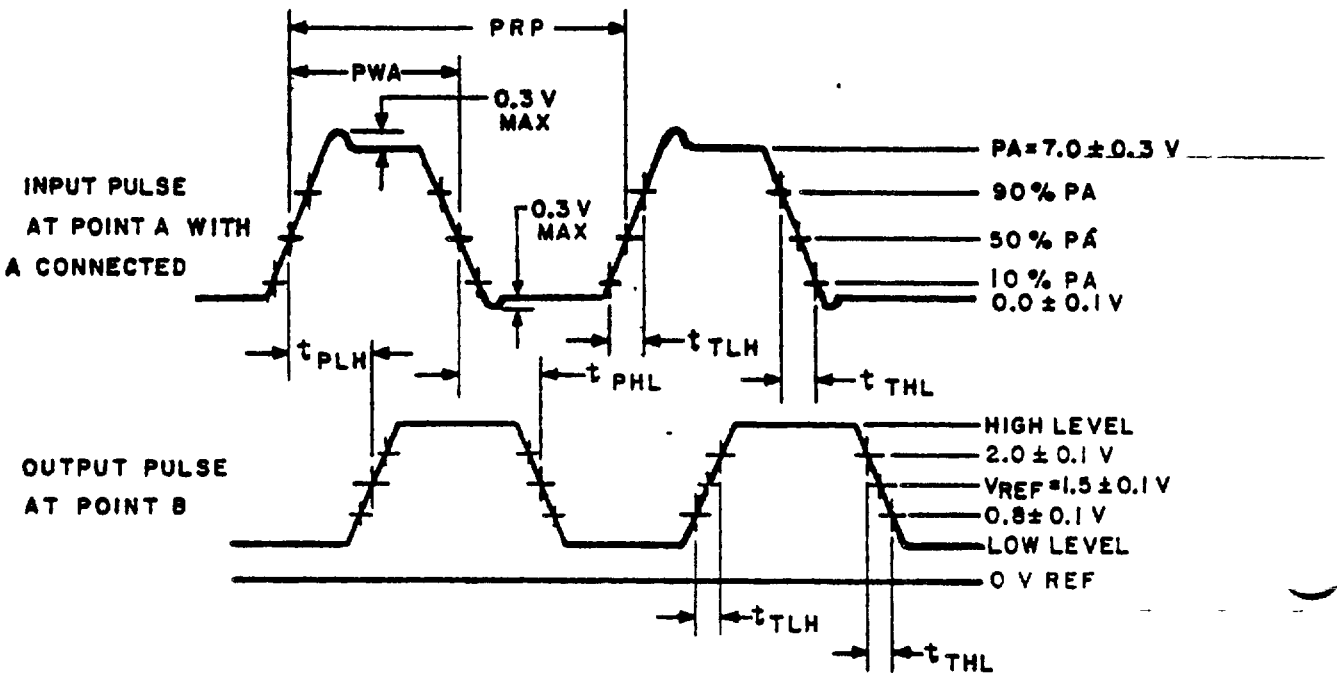
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NOTE: R1 = 51Ω ±5%, 1 W.

* FIGURE 21. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for table XX.

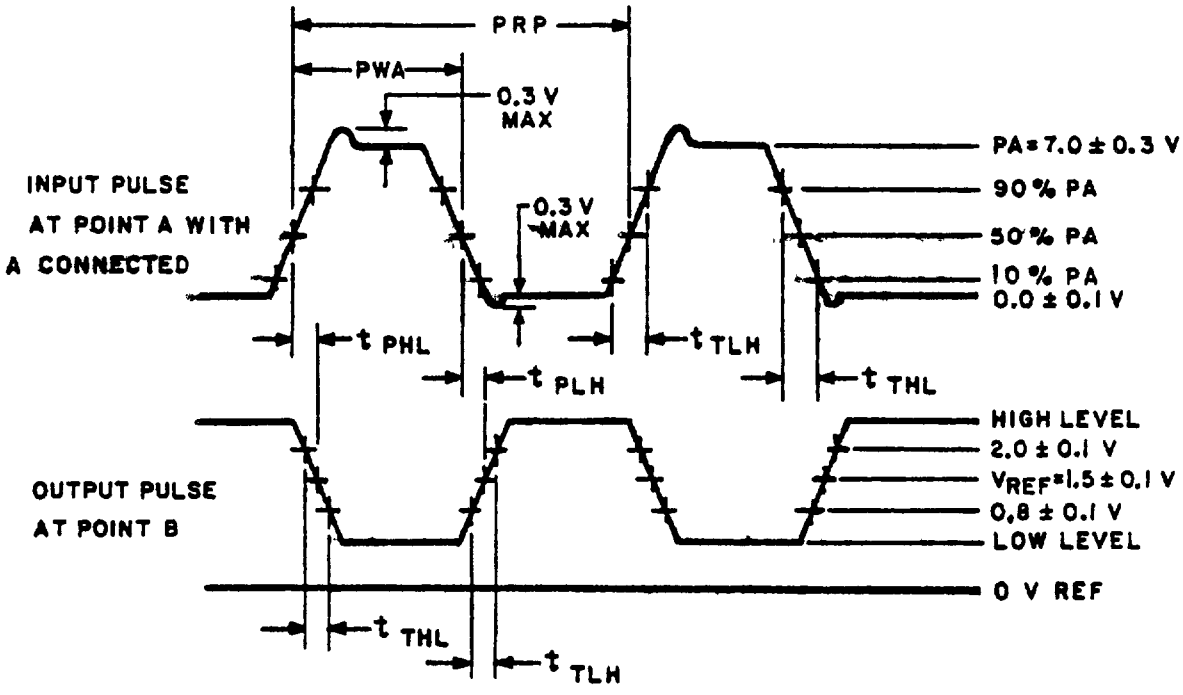
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NOTES:
1. Input pulse parameters:
   PWA = 5.0 ±0.5 µs.
   PRP = 10 ±1 µs.
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
2. Verify all outputs have a high level $\geq$ 2.4 V and low level $\leq$ 0.4 V before making measurements.
3. All voltages are referenced to 0 V unless otherwise specified.

* FIGURE 22. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XX.

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NOTES:
1. Input pulse parameters:
   PWA = 5.0 ±0.5 µs.
   PRP = 10 ±1 µs.
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
2. Verify all outputs have a high level $\geq$ 2.4 V and low level $\leq$ 0.4 V before making measurements.
3. All voltages are referenced to 0 V unless otherwise specified.

* FIGURE 23. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XX.

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* TABLE XXI. Propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter		Connect						Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
Y1	1E to 1H	Propagation delay time (t_{PLH}), C_L = 50 pF	Open	17	7	Open	8	8	3	25	ns	24,26
Y2	2E to 2H	"	"	13	12	"	9	9	"	"	"	"
Y3	3E to 3H	"	"	34	28	"	25	25	"	"	"	"
Y4	4E to 4H	"	"	35	29	"	37	37	"	"	"	"
Y5	1E to 1G	"	"	17	7	15	8	15	6	45	"	24,25
Y6	2E to 2G	"	"	13	12	14	9	14	"	"	"	"
Y7	3E to 3G	"	"	34	28	31	25	31	"	"	"	"
Y8	4E to 4G	"	"	35	29	32	37	32	"	"	"	"
Y9	1F to 1G	"	7	Open	17	15	Open	15	3	25	"	24,26
Y10	2F to 2G	"	12	"	13	14	"	14	"	"	"	"
Y11	3F to 3G	"	28	"	34	31	"	31	"	"	"	"
Y12	4F to 4G	"	29	"	35	32	"	32	"	"	"	"
Y13	1E to 1H	Propagation delay time (t_{PLH}), C_L = 150 pF	Open	17	7	Open	8	8	5	42	"	"
Y14	2E to 2H	"	"	13	12	"	9	9	"	"	"	"
Y15	3E to 3H	"	"	34	28	"	25	25	"	"	"	"
Y16	4E to 4H	"	"	35	29	"	37	37	"	"	"	"
Y17	1E to 1G	"	"	17	7	15	8	15	10	75	"	24,25
Y18	2E to 2G	"	"	13	12	14	9	14	"	"	"	"
Y19	3E to 3G	"	"	34	28	31	25	31	"	"	"	"
Y20	4F to 4G	"	"	35	29	32	37	32	"	"	"	"

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* TABLE XXI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

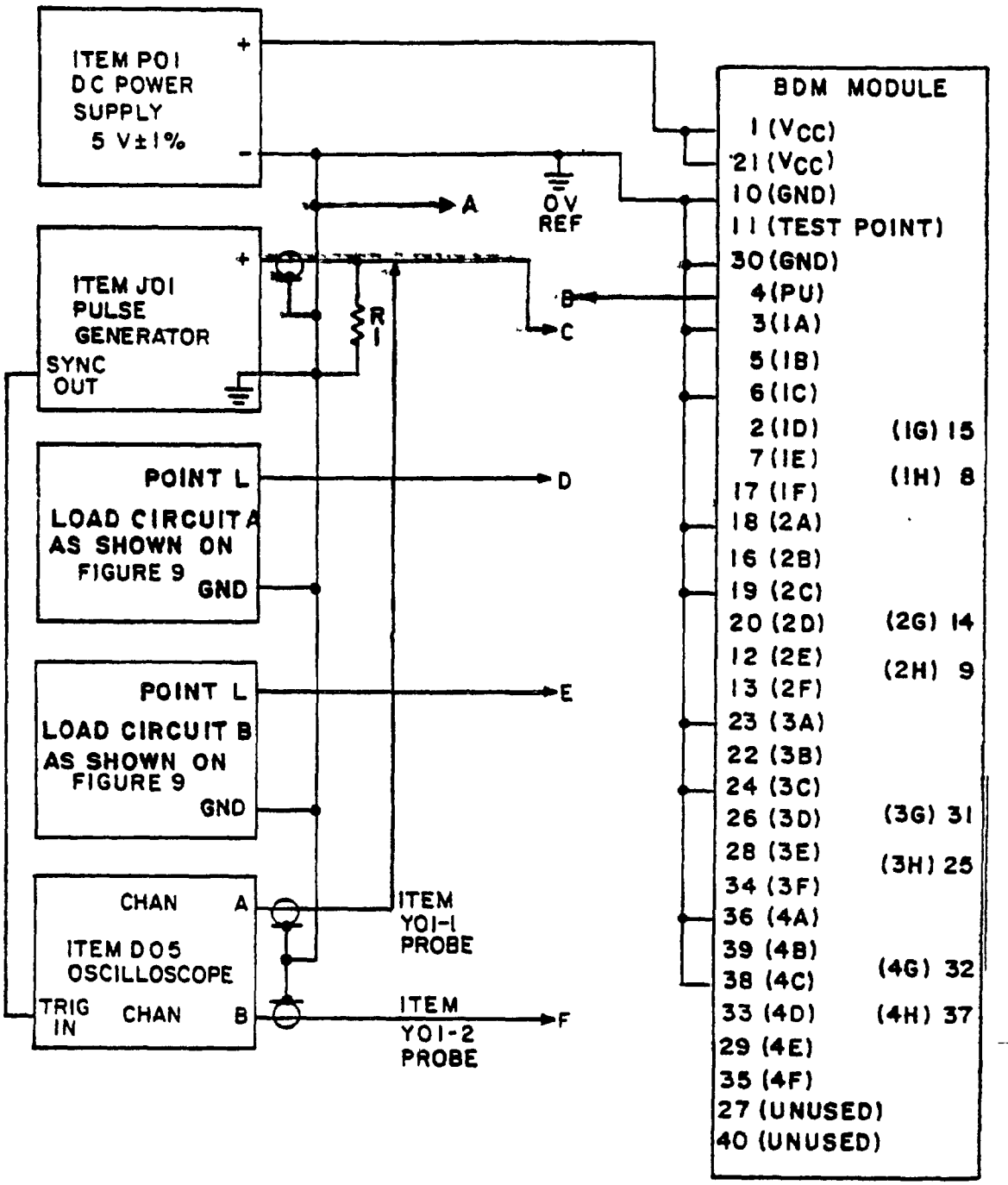
Test condition	Parameter		Connect						Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
Y21	1F to 1G	Propagation delay time (t_{PLH}), C_L = 150 pF	7	Open	17	15	Open	15	5	42	ns	24,26
Y22	2F to 2G	"	12	"	13	14	"	14	"	"	"	"
Y23	3F to 3G	"	28	"	34	31	"	31	"	"	"	"
Y24	4F to 4G	"	29	"	35	32	"	32	"	"	"	"
Z1	1E to 1H	Propagation delay time (t_{PHL}), C_L = 50 pF	Open	17	7	Open	8	8	3	27	"	"
Z2	2E to 2H	"	"	13	12	"	9	9	"	"	"	"
Z3	3E to 3H	"	"	34	28	"	25	25	"	"	"	"
Z4	4E to 4H	"	"	35	29	"	37	37	"	"	"	"
Z5	1E to 1G	"	"	17	7	15	8	15	6	52	"	24,25
Z6	2E to 2G	"	"	13	12	14	9	14	"	"	"	"
Z7	3E to 3G	"	"	34	28	31	25	31	"	"	"	"
Z8	4E to 4G	"	"	35	29	32	37	32	"	"	"	"
Z9	1F to 1G	"	7	Open	17	15	Open	15	3	20	"	24,26
Z10	2F to 2G	"	12	"	13	14	"	14	"	"	"	"
Z11	3F to 3G	"	28	"	34	31	"	31	"	"	"	"
Z12	4F to 4G	"	29	"	35	32	"	32	"	"	"	"
Z13	1E to 1H	Propagation delay time (t_{PHL}), C_L = 150 pF	Open	17	7	Open	8	8	5	40	"	"
Z14	2E to 2H	"	"	13	12	"	9	9	"	"	"	"
Z15	3E to 3H	"	"	34	28	"	25	25	"	"	"	"
Z16	4E to 4H	"	"	35	29	"	37	37	"	"	"	"

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* TABLE XXI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect						Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
Z17	1E to 1G	Propagation delay time (t_{PHL}), C_L = 150 pF	Open	17	7	15	8	15	10	82	ns	24,25
Z18	2E to 2G	"	"	13	12	14	9	14	"	"	"	"
Z19	3E to 3G	"	"	34	28	31	25	31	"	"	"	"
Z20	4E to 4G	"	"	35	29	32	37	32	"	"	"	"
Z21	1F to 1G	"	7	Open	17	15	Open	15	5	33	"	24,26
Z22	2F to 2G	"	12	"	13	14	"	14	"	"	"	"
Z23	3F to 3G	"	28	"	34	31	"	31	"	"	"	"
Z24	4F to 4G	"	29	"	35	32	"	32	"	"	"	"

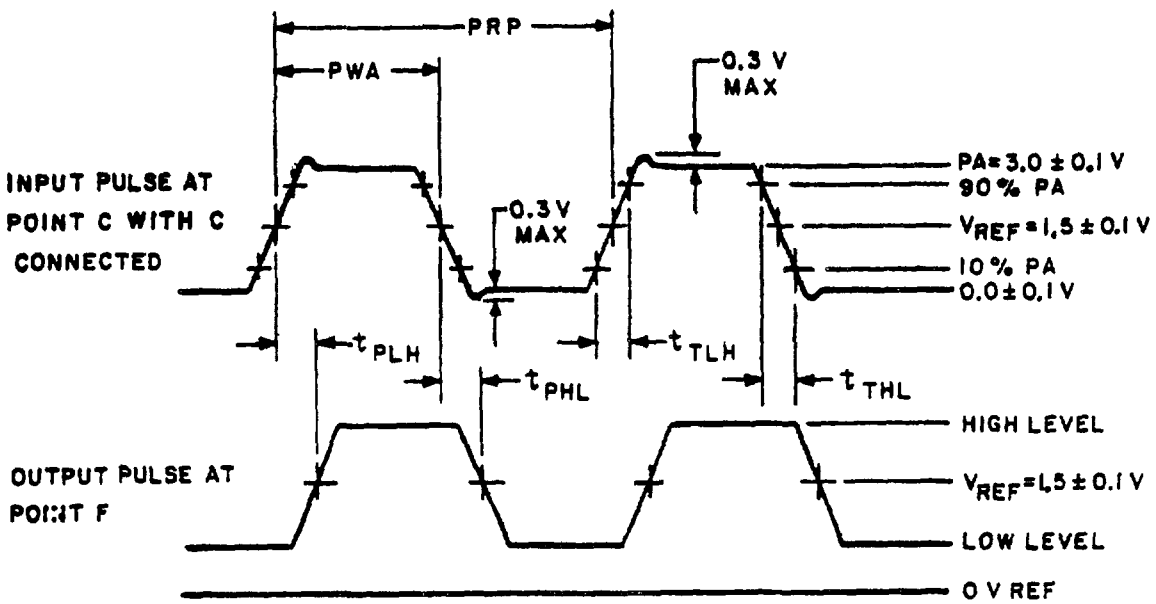
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NOTE: R1 = 51Ω ±5%, 1 W.

* FIGURE 24. Propagation delay time (t_{PLH} and t_{PHL}) test circuit for table XXI.

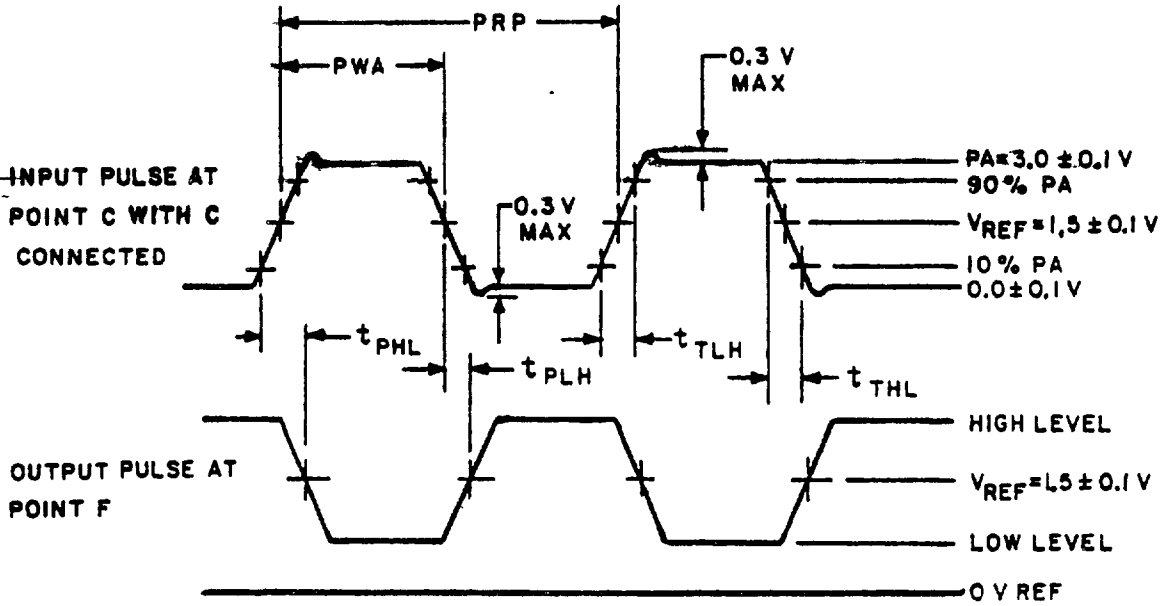
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NOTES:
1. Input pulse parameters:
   PWA = 5.0 ±0.5 µs.
   PRP = 10 ±1 µs.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Verify all outputs have a high level of ≥ 2.4 V and low level ≤ 0.4 V before making measurements.
3. All voltages are referenced to 0 V unless otherwise specified.

* FIGURE 25. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XXI.

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NOTES:
1. Input pulse parameters:
   PWA = 5.0 ±0.5 µs.
   PRP = 10 ±1 µs.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Verify all outputs have a high level of ≥ 2.4 V and low level ≤ 0.4 V before making measurements.
3. All voltages are referenced to 0 V unless otherwise specified.

* FIGURE 26. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XXI.

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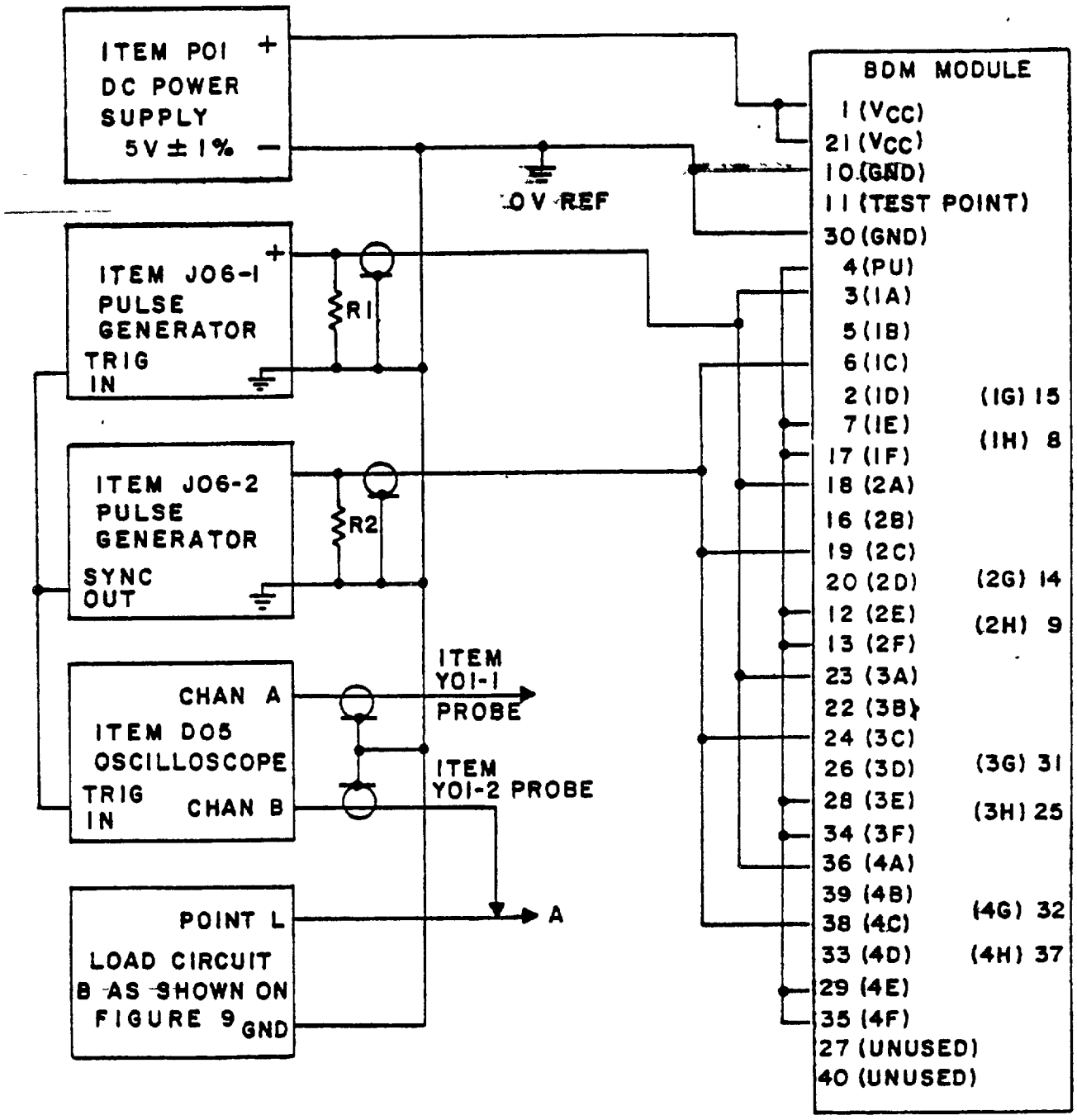
TABLE XXII. Common mode transient immunity (high-level) test.

Test condition	Parameter		Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
AA1	1H	Common mode transient immunity, high level	8	0	400	mV	27,28
AA2	2H	"	9	"	"	"	"
AA3	3H	"	25	"	"	"	"
AA4	4H	"	37	"	"	"	"

TABLE XXIII. Common mode transient immunity (low-level) test.

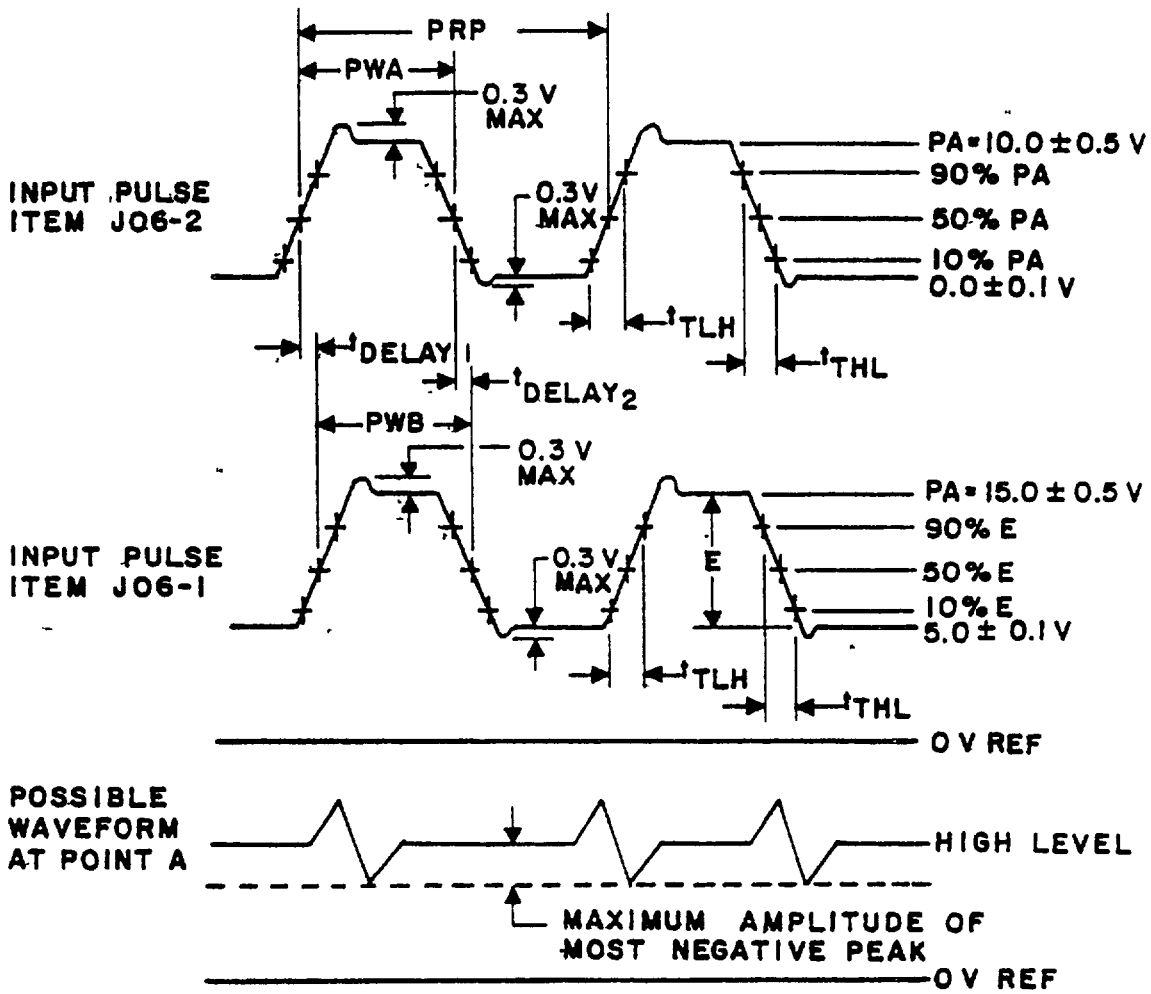
Test condition	Parameter		Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
BB1	1H	Common mode transient immunity, low level	8	0	400	mV	29,30
BB2	2H	"	9	"	"	"	"
BB3	3H	"	25	"	"	"	"
BB4	4H	"	37	"	"	"	"

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* FIGURE 27. Common mode transient immunity (high-level) test circuit for table XXII.

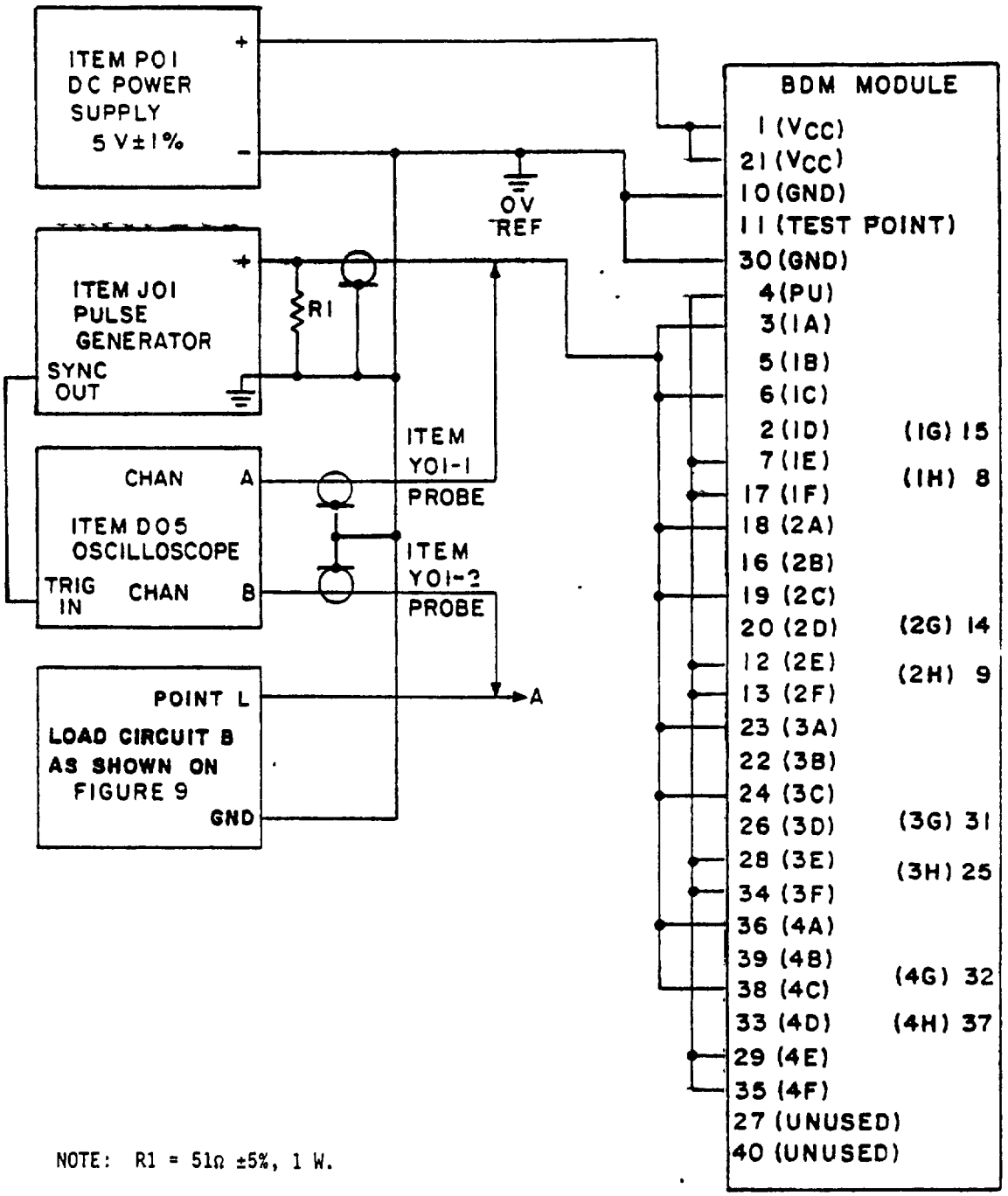
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NOTE: Input pulse parameters:
PWA = PWB = 5 ±1 µs.
PRP = 10 ±1 µs.
$t_{TLH} = t_{THL}$ = 100 +10, -0 ns.
$t_{DELAY1} = t_{DELAY2}$ = 0 ±5 ns.

* FIGURE 28. Common mode transient immunity (high-level) test circuit waveforms for table XXII.

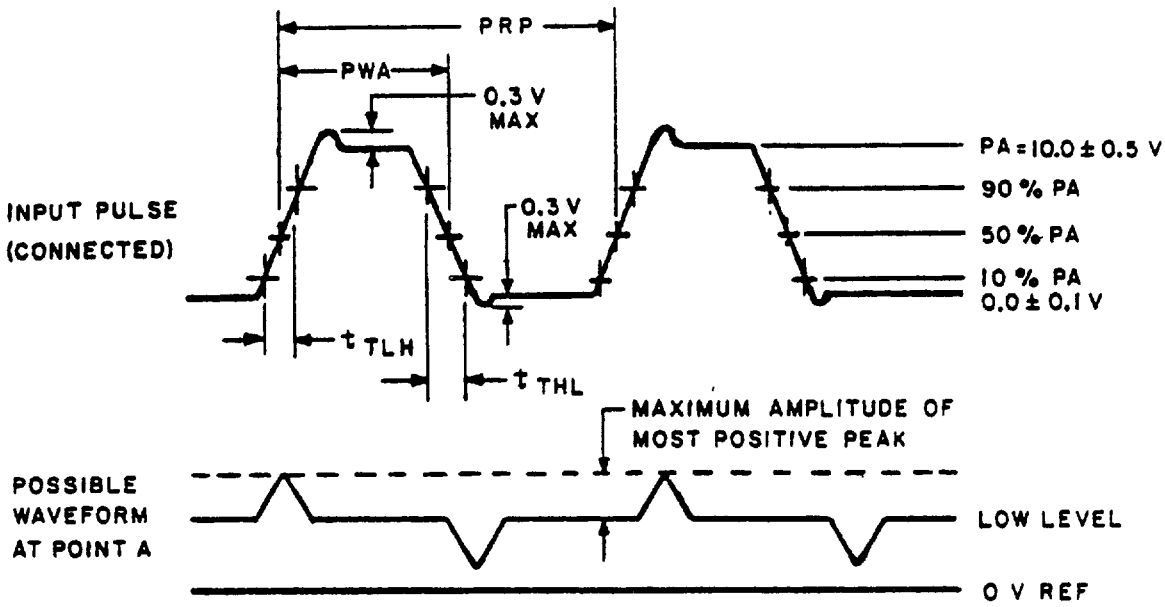
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NOTE: R1 = 51Ω ±5%, 1 W.

* FIGURE 29. Common mode transient immunity (low-level) test circuit for table XXIII.

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NOTE: Input pulse parameters:
PWA = 5 ±1 µs.
PRP = 10 ±1 µs.
$t_{TLH} = t_{THL} = 300\ ^{+30}_{-0}$ ns.

* FIGURE 30. Common mode transient immunity (low-level) test circuit waveforms for table XXIII.

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TABLE XXIV. Output crosstalk (high-level) test.

Test condition	Parameter		Connect						Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
CC1	1H	Output crosstalk, high level	3	7,12, 13,17, 28,29, 34,35	Open	Open	18,23, 36	8	0	400	mV	31,32
CC2	1H	"	Open	12,17, 28,29	3	7,13, 34,35	18,23, 36	8	"	"	"	"
CC3	2H	"	18	7,12, 13,17, 28,29, 34,35	Open	Open	3,23, 36	9	"	"	"	"
CC4	2H	"	Open	7,13, 28,29	18	12,17, 34,35	3,23, 36	9	"	"	"	"
CC5	3H	"	23	7,12, 13,17, 28,29, 34,35	Open	Open	3,18, 36	25	"	"	"	"
CC6	3H	"	Open	7,12, 29,34	23	13,17, 28,35	3,18, 36	25	"	"	"	"
CC7	4H	"	36	7,12, 13,17, 28,29, 34,35	Open	Open	3,18, 23	37	"	"	"	"
CC8	4H	"	Open	7,12, 28,35	36	13,17, 29,34	3,18, 23	37	"	"	"	"
CC9	1G	"	Open	7,12, 13,17, 28,29, 34,35	3	Open	18,23, 36	15	"	"	"	"
CC10	1G	"	3	7,12, 28,29	Open	13,17, 34,35	18,23, 36	15	"	"	"	"

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TABLE XXIV. Output crosstalk (high-level) test - Continued.

Test condition	Parameter		Connect						Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
CC11	2G	Output crosstalk, high level	Open	7,12, 13,17, 28,29, 34,35	18	Open	3,23, 36	14	0	400	mV	31,32
CC12	2G	"	18	7,12, 28,29	Open	13,17, 34,35	3,23, 36	14	"	"	"	"
CC13	3G	"	Open	7,12, 13,17, 28,29, 34,35	23	Open	3,18, 36	31	"	"	"	"
CC14	3G	"	23	7,12, 28,29	Open	13,17, 34,35	3,18, 36	31	"	"	"	"
CC15	4G	"	Open	7,12, 13,17, 28,29, 34,35	36	Open	3,18, 23	32	"	"	"	"
CC16	4G	"	36	7,12, 28,29	Open	13,17, 34,35	3,18, 23	32	"	"	"	"

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TABLE XXV. Output crosstalk (low-level) test.

Test condition	Parameter		Connect						Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
DD1	1H	Output crosstalk, low level	Open	7,12, 13,17, 28,29, 34,35	3	Open	18,23, 36	8	0	400	mV	31,32
DD2	1H	"	"	7,12, 17,28, 29	3	13,34, 35	18,23, 36	8	"	"	"	"
DD3	2H	"	"	7,12, 13,17, 28,29, 34,35	18	Open	3,23, 36	9	"	"	"	"
DD4	2H	"	"	7,12, 13,28, 29	18	17,34, 35	3,23, 36	9	"	"	"	"
DD5	3H	"	"	7,12, 13,17, 28,29, 34,35	23	Open	3,18, 36	25	"	"	"	"
DD6	3H	"	"	7,12, 28,29, 34	23	13,17, 35	3,18, 36	25	"	"	"	"
DD7	4H	"	"	7,12, 13,17, 28,29, 34,35	36	Open	3,18, 23	37	"	"	"	"
DD8	4H	"	"	7,12, 28,29, 35	36	13,17, 34	3,18, 23	37	"	"	"	"
DD9	1G	"	"	12,13, 17,28, 29,34, 35	3	7	18,23, 36	15	"	"	"	"
DD10	1G	"	"	12,17, 28,29	3	7,13, 34,35	18,23, 36	15	"	"	"	"

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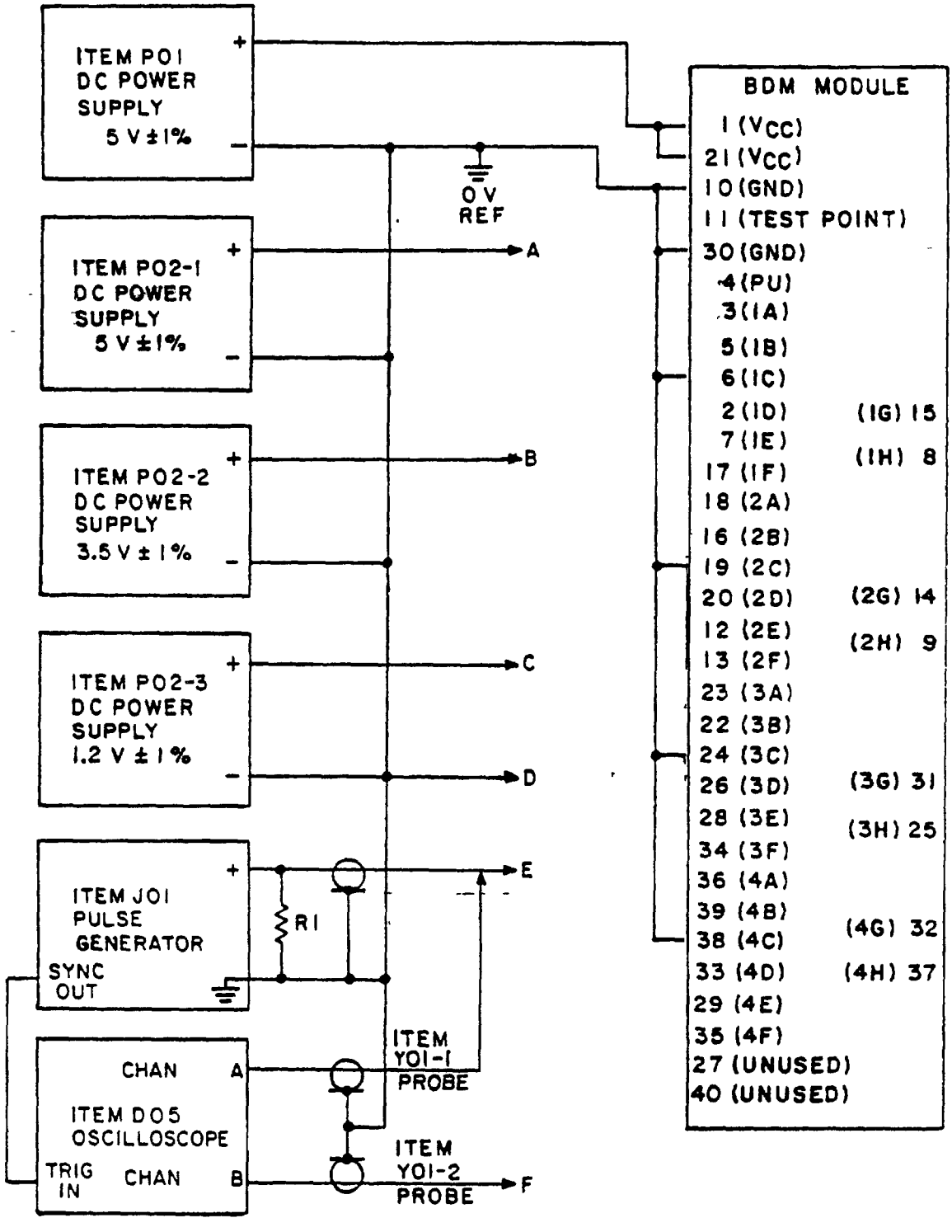
TABLE XXV. Output crosstalk (low-level) test - Continued.

Test condition	Parameter		Connect						Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
DD11	2G	Output crosstalk, low level	Open	7,13, 17,28, 29,34, 35	18	12	3,23, 36	14	0	400	mV	31,32
DD12	2G	"	"	7,13, 28,29	18	12,17, 34,35	3,23, 36	14	"	"	"	"
DD13	3G	"	"	7,12, 13,17, 29,34, 35	23	28	3,18, 36	31	"	"	"	"
DD14	3G	"	"	7,12, 29,34	23	13,17, 28,35	3,18, 36	31	"	"	"	"
DD15	4G	"	"	7,12, 13,17, 28,34, 35	36	29	3,18, 23	32	"	"	"	"
DD16	4G	"	"	7,12, 28,35	36	13,17, 29,34	3,18, 23	32	"	"	"	"

TABLE XXVI. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
EE	Power supply transient amplitude	0	400	mV	33,34

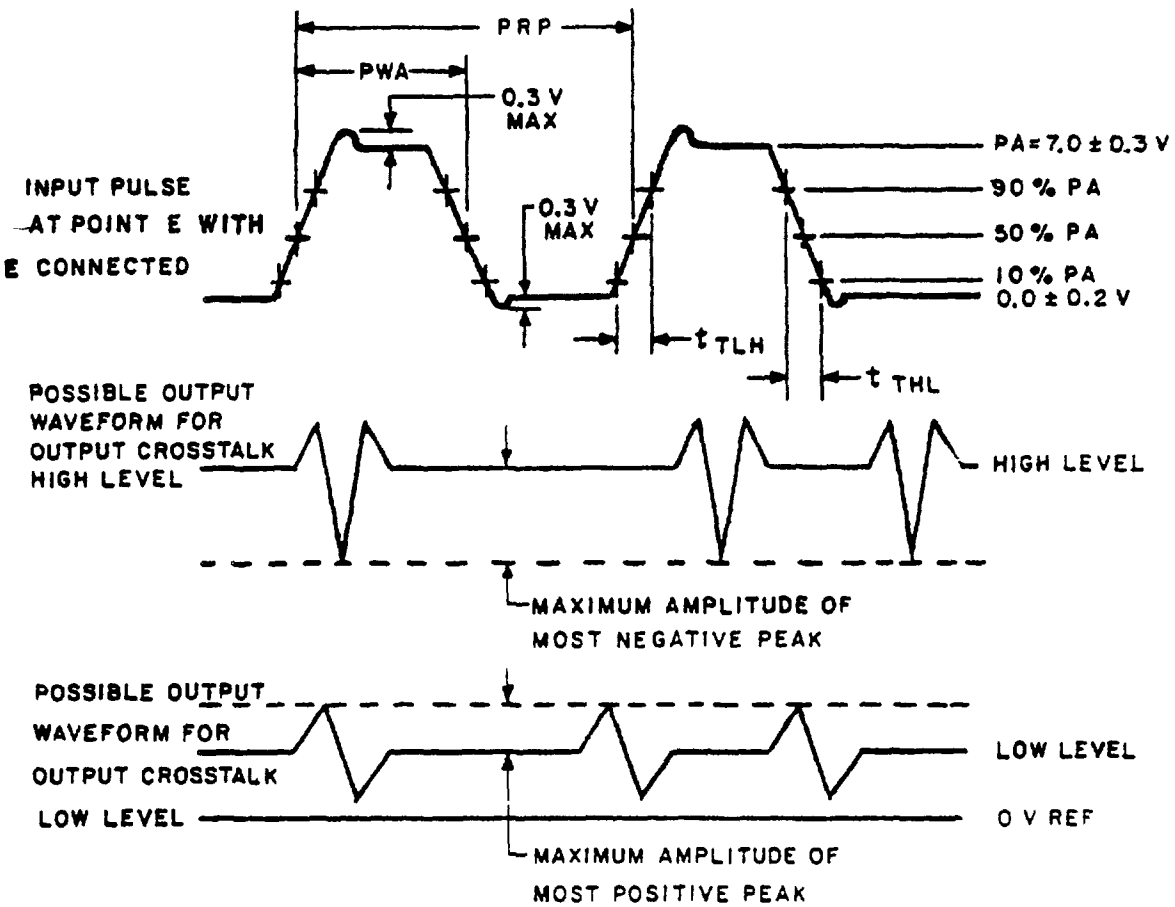
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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. Load all outputs with logic load as shown on figure 9 simultaneously. Load A to pins 15, 14, 31, and 32 and load B to pins 8, 9, 25, and 37.

FIGURE 31. Output crosstalk (high-level and low-level) test circuit for tables XXIV and XXV.

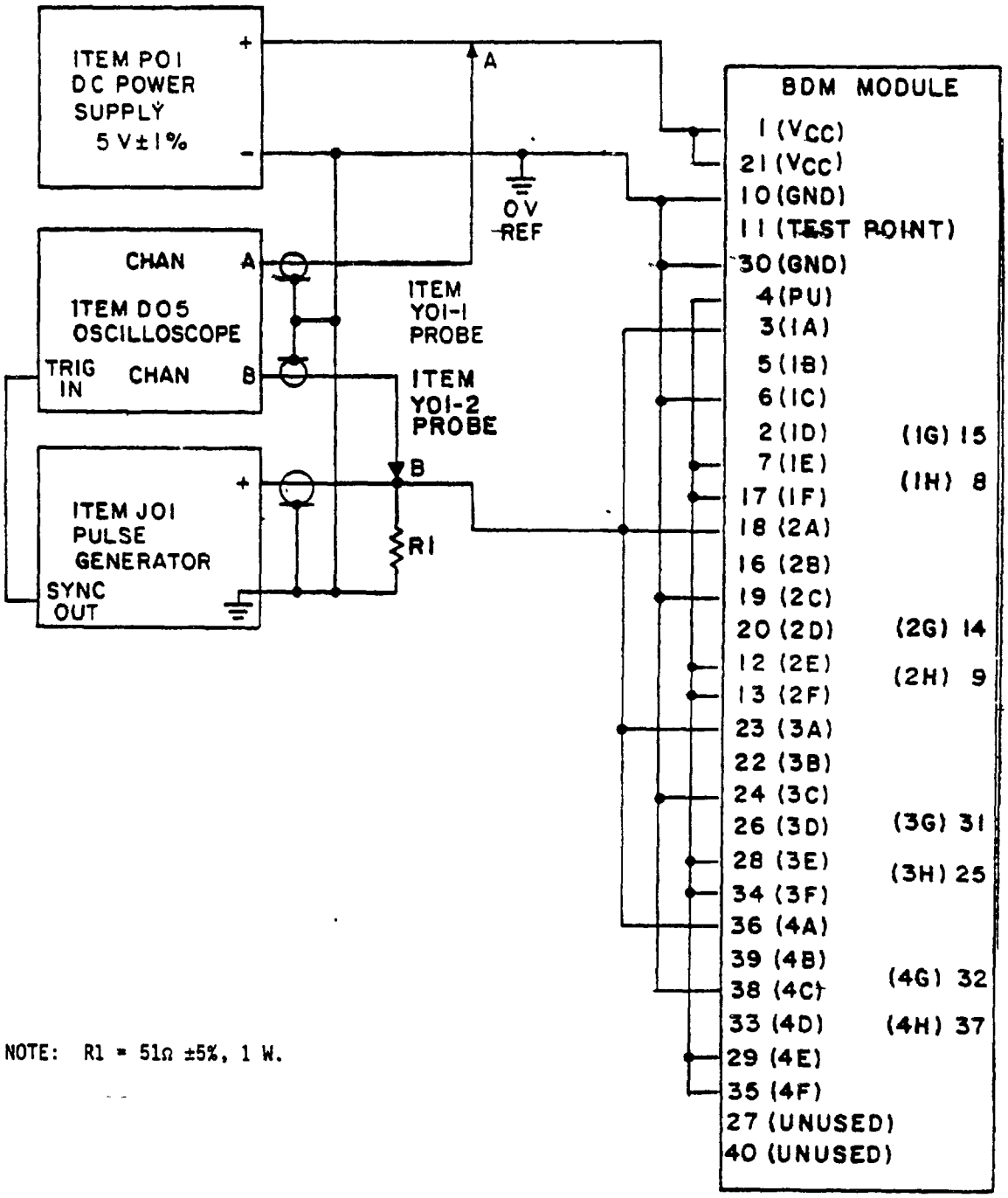
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NOTES:
1. Input pulse parameters:
   PWA 5 ±1 µs.
   PRP = 10 ±1 µs.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Lead length from load circuit to module output pin under test is to be less than 1 inch = 25.4 mm.

* FIGURE 32. Output crosstalk (high-level and low-level) test circuit waveforms for tables XXIV and XXV.

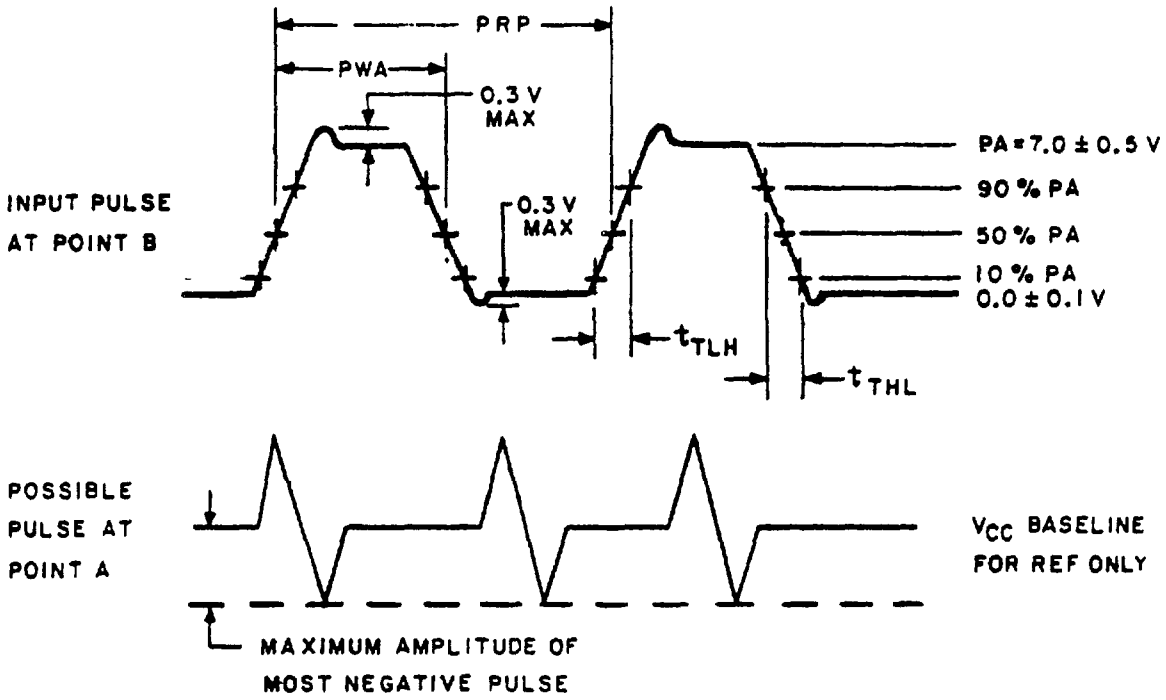
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NOTE: R1 = 51Ω ±5%, 1 W.

* FIGURE 33. Power supply transient amplitude test circuit for table XXVI.

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NOTES:
1. Input pulse parameters:
   PWA = 5.0 ±0.5 µs.
   PRP = 10 ±1 µs.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. All voltages are referenced to 0 V unless otherwise specified.

FIGURE 34. Power supply transient amplitude test circuit waveforms for table XXVI.

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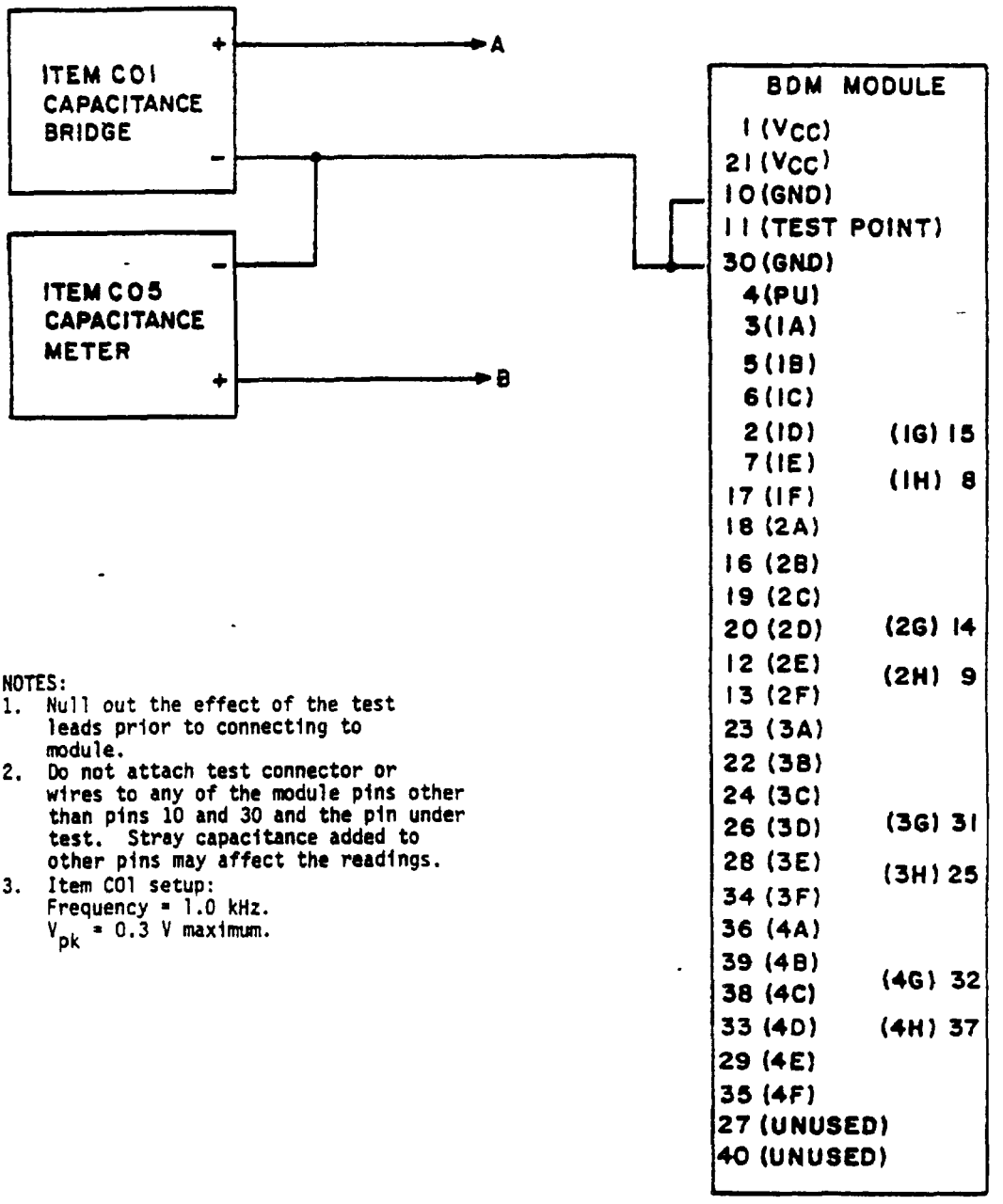
TABLE XXVII. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
FF	Power supply filter capacitance	1	Open	0.032	0.062	µF	35

TABLE XXVIII. Input capacitance test.

Test condition		Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
GG1	1E	Input capacitance	Open	7	2	15	pF	35
GG2	1F	"	"	17	"	"	"	"
GG3	2E	"	"	12	"	"	"	"
GG4	2F	"	"	13	"	"	"	"
GG5	3E	"	"	28	"	"	"	"
GG6	3F	"	"	34	"	"	"	"
GG7	4E	"	"	29	"	"	"	"
GG8	4F	"	"	35	"	"	"	"

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NOTES:
1. Null out the effect of the test leads prior to connecting to module.
2. Do not attach test connector or wires to any of the module pins other than pins 10 and 30 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Item C01 setup:
   Frequency = 1.0 kHz.
   V_{pk} = 0.3 V maximum.

* FIGURE 35. Power supply filter and input capacitance test circuit for tables XXVII and XXVIII.

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c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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